Internship Report

Climate, Land Use and Ecosystem Services (CLUES) M2 Master - Agrosciences, Environment, Territories, Landscape and Forest (AETPF) major.

Assessing how agricultural practices and landscape composition shape differences in arthropod biodiversity among habitat types in Europe using the PREDICTS database

Prepared by

Catalina Vattuone Troncoso

Co supervised by

Lucile Muneret, Nicolas Guilpart
UMR Agronomie, INRAE, Palaiseau, France; and AgroParisTech, Université Paris-Saclay, Palaiseau, France

and Rémi Duflot
Boreal Ecosystems Research Group, University of Jyväskylä, Jyväskylä, Finland

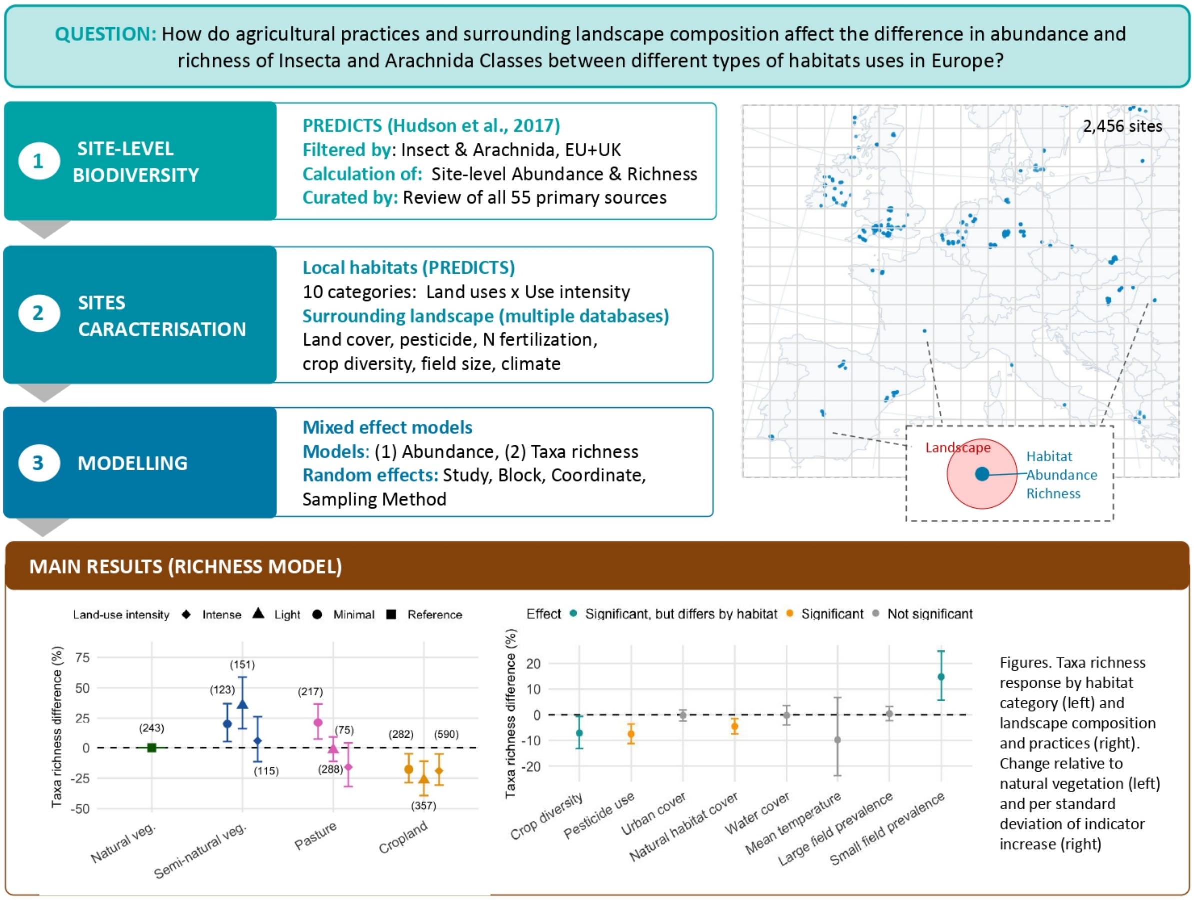
August 20th, 2026

Table of Contents

List of Figures

List of Tables

Abstract



Agriculture is often described as a main driver of arthropod decline, but the contribution of the individual practices behind it has not been disentangled at large scales. Analyses at that scale represent agriculture through broad land-use classes and generally reduce management intensity to a single proxy. This is partly because the global spatial data describing practices are coarse or absent. Another challenge is that available biodiversity records are compiled from many heterogeneous sources. In Europe, data describing agricultural inputs and structure of the agricultural landscapes is available at a finer resolution than their global equivalents. This study asked whether a global biodiversity database combined with independent European agronomic datasets can be used to separate the effects of individual practices and of the landscape composition on arthropods in Europe, and whether those effects differ between habitat types.

Records of Insecta and Arachnida for the European Union and the United Kingdom were extracted from the 2023 release of PREDICTS and curated by reference against the primary publications. The dataset comprised 235,661 records from 60 studies covering 2,456 sites sampled between 2000 and 2011. Each site was assigned a local habitat category crossing land-use classes with use intensity (from PREDICTS), and was characterized within a radius of 1 km by main land cover categories, pesticide and nitrogen application, crop diversity, field size and climate, drawn from public spatial products. The response of total abundance and taxa richness were then modelled with mixed-effects models.

Habitats under little or no management held higher taxa richness than intensively managed ones, with abundance following the same direction but with weaker support. The least disturbed habitat was not the richest: minimally used pasture held 96% more individuals and 21% more taxa than natural vegetation and lightly used semi-natural vegetation 72% and 36% more. A clear gradient across use intensity appeared only in pastures. Increased pesticide use in the surrounding landscape reduced abundance by 12.9% and richness by 7.5%, with no evidence that the effect differed between habitats, so semi-natural habitat are as much impacted as crop fields. The effects of crop diversity and the prevalence of small fields depended on local habitat, with greater crop diversity associated with higher abundance and richness in intensively used cropland, but lower in natural vegetation. Contrary to what was expected, natural vegetation cover in the landscape did not buffer these pressures. Marginal R^2 was low throughout (0.034 to 0.051), a pattern common to studies using this database that nonetheless calls for testing alternative modelling approaches.

Nitrogen application could not be separated from pesticides, and no spatial products were available for practices such as tillage, or organic management. Separating individual practices at this scale remains limited by the agronomic data, and by the biases of the biodiversity database. The analysis nonetheless provides, to our knowledge, the first estimate of the effect of pesticide application on the abundance and richness of insects and arachnids across habitat types at the European scale, together with a curated dataset that can be used in further work.

Institutional context

The work presented in this document was carried out during a 6-month research internship at **UMR Agronomie**, a joint research unit of INRAE**,** AgroParisTech and Université Paris-Saclay. The unit works on supporting the agroecological transition, with research organized around the production and mobilization of resources to help stakeholders innovate according to local contexts at the territorial and sectoral levels; improve the understanding of the ecological function of cultivated fields and the role of biodiversity in providing ecosystems services across a variety of cropping systems; and the quantification of the impacts of climate change on the balance between yield and biodiversity at global scale.

More information on *https://agronomie.versailles-saclay.hub.inrae.fr/*

1. Introduction

1.1. Background

Terrestrial biodiversity is declining globally, and over the past decade the focus of that concern has shifted from species extinctions to population declines and range contractions, in both vertebrates and invertebrates (Dirzo et al., 2014; Ceballos et al., 2017). For insects specifically, the concern is not only the loss itself but also the ecological functions associated with them, pollination, pest regulation, decomposition and nutrient cycling services and that agricultural production depends on (Goulson, 2019). This has motivated ambitious commitments to halt biodiversity loss, from the Kunming-Montreal Global Biodiversity Framework to the EU Biodiversity Strategy for 2030, whose Target 4 requires that by 2030 habitats and species show no deterioration in conservation trends and status (European Commission, 2020).

Substantial gaps in understanding the decline nonetheless persist. For arthropods, the past six years have seen repeated debate over its magnitude, with estimates ranging from over 40% of species threatened with extinction within a few decades (Sánchez-Bayo and Wyckhuys, 2019) to an average decline in terrestrial insect abundance of around 9% per decade, alongside a very debated increase in freshwater insect abundance of around 11% per decade, from time series with a median start year of 1986 (van Klink et al., 2020), and each of these publications has been contested (Desquilbet et al., 2020; Wagner, 2019; Daskalova et al., 2021). Furthermore, Valdez et al. (2023) showed that reliable global trends in local species richness would require monitoring an unfeasibly large number of perfectly sampled sites. Uncertainty about the decline magnitude is not, however, a reason to postpone research into its causes. Harvey et al. (2020) set out research priorities that include comparing insect abundance and diversity across management-intensity gradients. And Halsch et al. (2025), after synthesing hypotheses from over 175 reviews, found agricultural intensification to be one of the most cited drivers of diversity decline, and highlighted the importance to continue its research.

1.2. The scale gap between local studies and global models

Evidence on how agriculture affects arthropods has accumulated at two scales, each with its own limitations. Global biodiversity models represent agriculture through broad land-use classes, sometimes divided into coarse use-intensity categories, so that the effects of land-use intensity are not yet adequately reflected in biodiversity scenarios (Dullinger et al., 2021; Alkemade et al., 2022). In practice, intensification is reduced to a single proxy, generally nitrogen fertilization, which is a central determinant of crop productivity but only one of the inputs distinguishing management systems (Lassaletta et al., 2014). Part of the difficulty of integrating other variables is that the data needed to describe these inputs spatially do not exist at the resolution the question demands. Even fertilizer application is generally available only at coarse spatial resolution (Dullinger et al., 2021). The pesticide maps available for Europe at are derived by downscaling global estimates and calibrating them against national sales statistics (Porta et al., 2025), and no comparable spatially explicit product is even available for practices such as tillage, or mechanical weeding.

Field studies work from the opposite direction. They measure the effect of a particular practice on a particular community with precision, but their results are limited to the context in which they were obtained. Smith et al. (2020), analysing 60 crop types across six continents, found that the biodiversity benefit of organic over conventional farming increased with the average crop field size of the surrounding landscape, so the same practice delivered different gains depending on where it was applied. Batáry et al. (2011), synthesising 109 observations for species richness and 114 for abundance, found that agri-environmental management raised richness in croplands in structurally simple landscapes but not in complex ones, while in grasslands the effect held regardless of landscape context. Kleijn et al. (2006), comparing 202 paired fields in five European countries, found that different agri-environment schemes benefit different arthropod group depending on the context. Field studies therefore build a large body of reliable evidence in which the effect of a particular practice depends on where it is measured.

1.3. Filling the gap

Large-scale systematic monitoring could, in the future, provide robust evidence for exploring these relationships across many conditions. In fact, efforts to create a coordinated monitoring body to report against biodiversity commitments are underway (Kissling and Lumbierres, 2023; Kissling et al., 2026). However, the system is only beginning to be built and the gaps it is intended to close remain wide (Santana et al., 2025).

In the meantime, global biodiversity databases compiled records from many independent sources into a common structure, allowing observations to be analyzed together across large geographic extents. **PREDICTS** assembles spatial comparisons of biodiversity across sites differing in land use and land-use intensity, with more than 3.2 million records from over 26,000 sites and 47,000 species in its original 2016 release (Hudson et al., 2017). **BioTIME 2.0** takes a temporal rather than spatial approach, compiling almost 12 million records spanning 1874 to 2023 (Dornelas et al., 2025), while **GBIF** aggregates close to three billion occurrence records from natural history collections, citizen science networks and environmental recording schemes (GBIF.org, 2026), which suit questions of where species occur but are mostly collected without a sampling design. Taxon-specific databases also exist, such as **InsectChange** which aims to track temporal change in insect and arachnid assemblages worldwide (van Klink et al., 2021).

These databases have supported a range of findings on agriculture and biodiversity. Using PREDICTS, Newbold et al. (2015) estimated that land use and related pressures had reduced average local species richness by 13.6% and total abundance by 10.7% globally, reaching 76.5% and 39.5% in the worst-affected habitats, with land-use intensity among the pressures with the strongest effect. Later work has refined this picture, showing that losses depend on the interaction between agriculture and climate warming (Outhwaite et al., 2022), on crop type and rotation (Fan et al., 2024), and on geography and the amount of natural habitat remaining (Ceaușu et al., 2025). For arthropods specifically, Scheepens et al. (2026) found agricultural impacts to be less negative in Europe than elsewhere, attributing this to a degraded baseline of primary vegetation, and Millard et al. (2021), using separate models for fertiliser application, found that insects responses differ among taxa. These analyses have not yet reached the point of disentangling the effects of individual drivers, however, and they share two further limitations. The predictors explain only a small part of the variation in the data (Appendix 1), and model fit is reported inconsistently or not at all.

Assembling hundreds of independent studies into a single database is also error-prone, and serious problems have been documented. Gaume and Desquilbet (2024), reviewing the sources behind InsectChange, identified 553 issues across 17 problem types affecting 161 of its 165 datasets, from errors in insect counts to the inclusion of experimentally manipulated studies and inadequate coordinates. To our knowledge no equivalent review exists for the other databases of this type, including PREDICTS, on which the findings above were based.

1.4. Aim of this study

The goal of this research is, therefore, to study whether we can use a global biodiversity database to further understand the effect of individual agricultural practices on arthropods by compiling a dataset based on the PREDICTS database and on independent large-scale agronomic datasets (Table 2-2)The main question we addressed is:

How do agricultural practices and surrounding landscape composition affect the difference in abundance and richness of Insecta and Arachnida Classes between different types of habitats uses in Europe?

To answer this question, a site-level dataset of Insecta and Arachnida records was extracted from PREDICTS for the European Union and the United Kingdom and screened for issues. Each site was characterised by its local land use and management intensity, and by indicators of agricultural practices and landscape composition drawn from spatial datasets. Total abundance and richness were then modelled using mixed-effects models.

The five hypotheses of the work are represented graphically in Figure 1-1 and described below.

At the local scale, we expect:

I. natural unmanaged habitats (e.g., natural and semi natural vegetation of light or minimal use intensity) to host higher abundance and richness than less natural, more intensively managed habitats (e.g., cropland or pastures of intense use);

II. an increase in local management intensity has greater impact in less natural than in more natural habitats.

At the landscape scale, we expect:

III. landscapes with greater management intensity (higher pesticide, higher fertilization, less compositional heterogeneity and less configurational heterogeneity in the surrounding landscapes) host lower abundance and richness;

IV. the impact of landscape intensity to be more pronounced in natural local habitats than in already degraded intensively used habitats;

V. increase proportion of natural vegetation in the surrounding landscape to buffer the decline in abundance and richness caused by pressures.

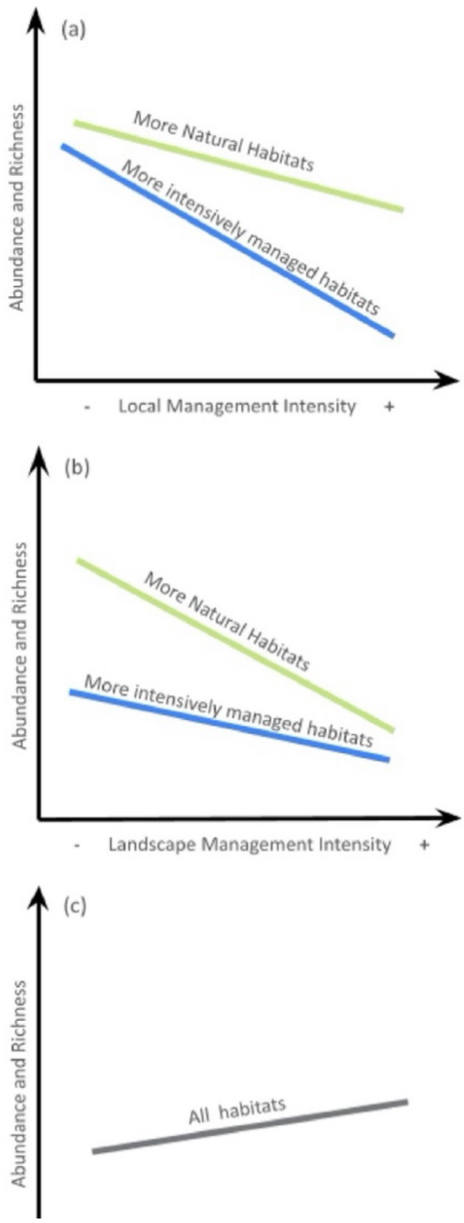


Figure 1-1. Schematic representation of the research hypotheses regarding the effect of local management intensity (a), landscape management intensity (b), and percentage of natural vegetation in surrounding landscape (c), on site-level abundance and richness.

2. Method

The analysis followed 3 steps. First, a site-level biodiversity dataset of Insecta and Arachnida was extracted from a global biodiversity database and curated against its primary references (Section 2.1). Second, each site was characterized, first locally by assigning a habitat category (Section 2.2) and then in its surroundings, by estimating landscape indicators through spatial datasets, including land cover, agricultural inputs, and the compositional and configurational heterogeneity of the agricultural mosaic (Section 2.3). Finally, the relationships between abundance and richness, local habitat and landscape variables were analyzed using mixed-effects models (Section 2.4).

All data processing, spatial extraction and statistical analysis were carried out in R version 4.5.2 (R Core Team, 2025). Main packages are listd in Appendix 2.

2.1. Biodiversity data

An in-depth review and characterisation of BioTIME 2.0 and PREDICTS was carried out to determine which global database was best suited for the analysis. Given its availability of local land use and use intensity information, and the superior number of studies and sites covering arthropods in Europe, the 2023 release of PREDICTS was selected. It contains 3,278,056 biodiversity records across all terrestrial taxa, from 666 studies, whose global distribution is shown in Figure 2-1.

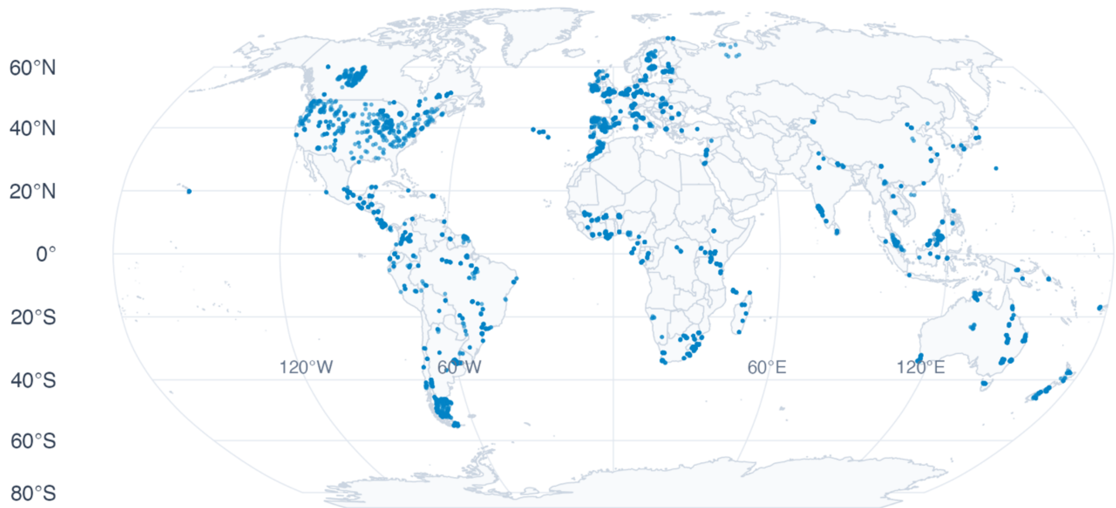


Figure 2-1. Global distribution of all sampling sites in PREDICTS database v.2023.

The structure of PREDICTS is represented in Figure 2-2. Records are organised by source, corresponding to an individual publication; by study, grouping records from a single sampling method or country within a source; and by site, identified by a unique SSBS code carrying coordinates, land use and use intensity, sampling dates, and sampling method and effort. Sites are sometimes grouped into blocks ("SSB" code), indicating locations spatially close to one another. Each site holds multiple observations, each with a taxonomic identification, a metric -abundance, richness or occurrence, with abundance accounting for 82.5% of all observations-, a raw measurement and an effort-corrected measurement (standardised within studies). Taxonomic resolution varies from species to class.

Because two or more sites with distinct SSBS codes can share the same coordinates, when a location was sampled with more than one method, a Coordinate_ID was created by concatenating latitude and longitude.

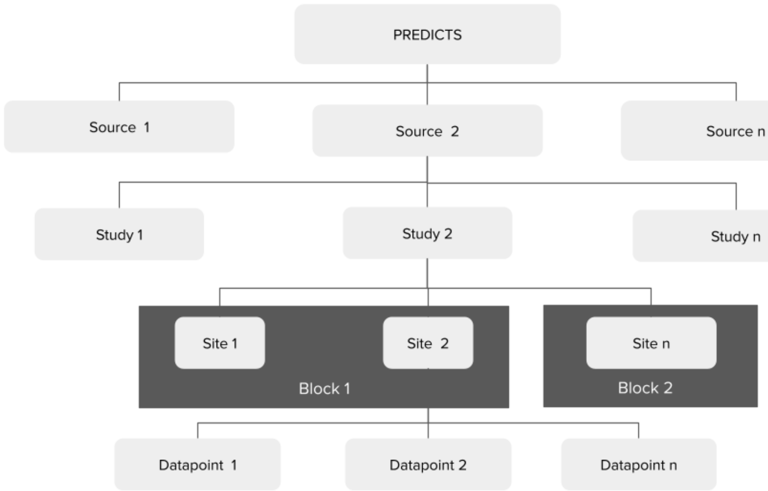


Figure 2-2. PREDICTS data structure.

For the purpose of this study, we selected the Insecta and Arachnida observations ("Class") recorded at sites in the continental area of the European Union and the United Kingdom ("Country"), for which the diversity metric corresponded to abundance of individuals ("Diversity_metric_type" and "Diversity_metric_unit"). Observations at sites whose land use was recorded as "Cannot decide" were also removed. The resulting initial set comprised observations from 55 references.

All primary references behind this set were then reviewed. Publication titles were retrieved from the PREDICTS list of resources, except for sources identified as unpublished data. A questionnaire designed to identify potential issues, in particular those raised by Gaume and Desquilbet (2024), was completed for each reference (Appendix 3 and full answers in Supplementary Information S3). Issues were not restricted to errors in the database, such as differences in counts, but also included features that impose constraints on the analysis, such as time series, or that limit the review itself, such as absence of raw data.

The results were systematised by source and issues were categorized. A decision was taken for each issue, ranging from reporting it, through excluding the source or a specific group of sites, to correcting the data on the basis of information provided by the original authors or standardising non-primary fields. The full list of issues is available in Supplementary Information S1, and a summary is presented in Appendix 4.

After management actions were taken, further selection was applied to retain only sites with predominant land use corresponding to primary vegetation, secondary vegetation of different ages, cropland and pastures, independent of their use intensity.

The resulting biodiversity dataset comprises 235,661 records from 38 sources, 60 studies and 18 countries, covering 2,456 sites (SSBS codes) at 1,405 unique coordinates (Figure 2-3). Sites were sampled between 2000 and 2011. The taxonomic and temporal distribution is available in Appendix 5.

A site-level dataset was subsequently produced by grouping the raw observations of the final dataset by SSBS. The following metrics were calculated for each site:

- **Total abundance (TA):** the sum of the raw abundance (individuals counts) of each taxon sampled at the site;
- **Taxa richness:** the number of all unique lowest-level taxa sampled at the site

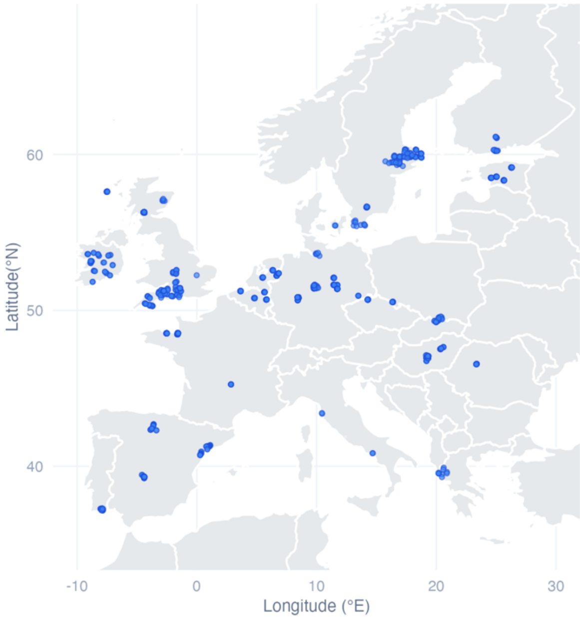


Figure 2-3. Distribution of sampling sites across the European Union and the United Kingdom after data curation. Points represent the unique coordinates retained in the final dataset extracted from the PREDICTS database v.2023.

Finally, two references found during the curation to be usable only for abundance were excluded from the richness analyses. The latest was performed with a slightly smaller dataset containing 235,566 records from 37 references and 2441 sites.

2.2. Local land use and use intensity categories

For each site, PREDICTS provides a land-use class, a use-intensity class, and a longer description of the habitat in which the sample was taken. Land use in our dataset is classified by PREDICTS as primary vegetation, secondary vegetation -divided into mature, intermediate and young, or indeterminate where the stage was unknown-, cropland, or pasture. Within each land use class, the land use intensity is classified as minimal, light or intense. Both were assigned by the PREDICTS compilers from the description given in the source publication or from information subsequently provided by the data contributors (Hudson et al., 2017).

A manual review was carried out to compare the habitat descriptions with the assigned land uses. This showed that the land-use class generally reflects the very specific local condition of the sample rather than the surrounding context. For example, samples taken at field margins, for instance, are assigned to young secondary vegetation rather than to cropland land use. The review also showed that primary vegetation sites frequently shared characteristics with mature secondary vegetation sites, and that grassland, while commonly assigned to pasture, were in some cases assigned to secondary vegetation, cropland or even primary vegetation. A list of the concerned studies is presented in Appendix 6.

Land uses were therefore regrouped into four classes:

- **Natural vegetation:** sites classified by PREDICTS as primary vegetation or mature secondary vegetation;
- **Semi-natural vegetation:** sites classified as intermediate secondary vegetation, young secondary vegetation, or secondary vegetation of indeterminate age;
- **Cropland:** sites classified as cropland;
- **Pasture:** sites classified as pasture.

Local habitat categories were then defined by crossing the four land-use classes with use intensity, giving ten categories (Table 2-1). Primary vegetation sites were all of minimal use, and mature secondary vegetation comprised 77 sites of minimal use and nine of light use; these were merged into a single natural vegetation category without a use-intensity level.

Table 2-1.Local habitat categories. Categories are defined by crossing the land-use classes with use intensity

Local Habitat Category	Abbreviation	Number of sites Abundance	Number of sites Richness
Natural vegetation	NV	243	243
Semi-natural vegetation minimal use	SNV-M	126	123
Semi-natural vegetation light use	SNV-L	151	151
Semi-natural vegetation intense use	SNV-I	115	115
Cropland minimal use	C-M	282	282
Cropland light use	C-L	369	357
Cropland intense use	C-I	590	590
Pasture minimal use	P-M	217	217
Pasture light use	P-L	288	288
Pasture intense use	P-I	75	75
Total		**2456**	**2441**

2.3. Landscape characterization data

The landscape surrounding each sampling site was characterised from publicly available spatial datasets, selected to represent potential variables affecting the biodiversity response in the targeted habitat categories. The variables, sources, resolutions and reference years are summarised in Table 2-2.

Table 2-2. Spatial datasets used to characterize the landscape surrounding each sampling site. Sources can have Global coverage (G), European coverage (E), or European Union and United Kingdom only coverage (EU + UK)

Category	Indicator	Source	Resolution	Time coverage
Land Cover	Percentage of cropland surface (%)	Liao et al. (2026) GACED30 (G)	30 mts	2000 to 2024 yearly
	Percentage of water surface (%)	EEA (2020) Copernicus HRL Water & Wetness 2018 (E)	100 mts	2012-2018 imagery
	Percentage of urban surface (%)	EEA (2020) CORINE Land Cover (E)	100 mts	2000, 2006, 2012, and 2018
	Percentage of natural vegetation surface (%)	EEA (2020) CORINE Land Cover (E)	100 mts	2000, 2006, 2012, and 2018
	Percentage of pasture surface (%)	EEA (2020) CORINE Land Cover (E)	100 mts	2000, 2006, 2012, and 2018
Agricultural Inputs	Average total pesticide application rate (kg/ha)	Porta et al. (2025) (EU + UK)	250 mts	2015, 2020
	Average nitrogen fertilizer rate (kg/ha)	Nguyen et al. (2024) NPKGRIDS (G)	0,05°	2020
Compositional heterogeneity	Average Shannon Index	Tang et al. (2024) CROPGRIDS (G)	0,05°	2020
	Average Alpha Diversity of Crops	Machefer et al. (2024) (EU + UK)	1 km	2018
Configurational heterogeneity	Percentage of surface where small and very small fields dominate (%)	Lesiv et al. (2019) (G)	0,005°	2000-2017
	Percentage of surface where large and very large fields dominate (%)	Lesiv et al. (2019) (G)	0,005°	2000-2017
Climate	Mean temperature (°C)	Muñoz-Sabater et al. (2021) ERA5-Land re-analysis (G)	0,1°	1950 to today monthly
	Total annual precipitation (mm)	Muñoz-Sabater et al. (2021) ERA5-Land re-analysis (G)	0,1°	1950 to today monthly

Predictor values were extracted for each unique coordinate within a circular buffer of 1 kilometre, weighting each pixel of the spatial products listed in by the proportion of its area falling inside the buffer, so that pixels only partially covered contribute in proportion to the overlap. Buffer geometries were reprojected into the coordinate reference system of each raster. Where a dataset was available for more than one year, the closest year was matched to the sampling year at the start of the sampling events recorded for each site. Where only a single reference year was available, that layer was applied to all sites.

A collinearity assessment was later performed between the predictors to select only uncorrelated predictors for the statistical modeling (Section 2.4). Below is a brief description of each indicator, with more detailed information presented in Appendix 7.

2.3.1. Percentage of cropland

Cropland cover was obtained from the Global 30-m Annual Cropland Extent Dynamics dataset (GACED30; Liao et al., 2026), selected for its 30 m resolution, the finest available among global cropland products, and its annual coverage from 2000 to 2024. The product identifies cropland from evidence of active management rather than from the state of the surface at a single date and aligns its definition with the FAO cropland category. It includes annual herbaceous crops, perennial woody crops, land under active fallow and agricultural structures, and excludes temporary meadows and pastures, that is land cultivated with forage for less than five years. The variable is the percentage of the buffer classified as cropland the year when the sampling started.

2.3.2. Percentage of water surface

Inland water and wetland cover were obtained from the High Resolution Layer Water and Wetness of the Copernicus Land Monitoring Service (European Union's Copernicus Land Monitoring Service Information, 2020), a 100 m product for the 2018 reference period. The layer classifies each pixel by the frequency with which water or wet soil conditions are observed, independently of the vegetation growing on it. The four classes describing permanent and temporary water and wetness were combined into a single mask, and the variable is the percentage of the buffer covered by that mask. Wetlands are accounted here rather than in the natural vegetation variable, as they are accounted for by the temporary wet category. Sea water is coded separately in the source layer and was excluded from the indicator, but was extracted as a separate variable for the purpose of assessing total land cover, described in Section 2.3.3.

2.3.3. Percentage of natural vegetation, urban and pasture area

Urban, natural vegetation and pasture cover were obtained from CORINE Land Cover (European Union's Copernicus Land Monitoring Service Information, 2020b, 2020c), the 100 m European land cover product, in its 2020 revision. Two vintages were used, 2000 and 2006, each site being assigned the vintage closest to and not later than its sampling year; later vintages were not required because sampling in the dataset does not extend beyond 2011. Three variables were derived as the percentage of the buffer occupied by (1) urban surfaces, by (2) pastures, and by (3) the total natural vegetation, including classes describing forest, natural grassland, moors and heathland, sclerophyllous vegetation and transitional woodland-shrub. The natural vegetation excludes the classes describing open spaces with little or no vegetation, and the wetland class, which is accounted for in the water cover indicator. The full list of codes is given in Appendix 7.

Because the land cover variables come from three different sources, their shares do not necessarily sum to 100% within a buffer. Where the total exceeded 100%, the urban, pasture, natural vegetation and water covers were rescaled proportionally, while cropland and sea water were held fixed. The procedure is described in Appendix 7.

2.3.4. Climate conditions

Mean annual temperature and total annual precipitation were obtained from the ERA5-Land monthly averaged re-analysis (Muñoz-Sabater et al., 2021), produced by the European Centre for Medium-

Range Weather Forecasts at a resolution of 0.1°. For each site, the twelve monthly layers of its sampling year were used. Mean annual temperature was calculated as the mean of the monthly temperature layers, and total annual precipitation by converting the monthly daily-accumulation values to monthly totals and summing them across the year.

2.3.5. Pesticides

Pesticide application rates were obtained from the European pesticide application rate maps of Porta et al. (2025), which cover the member states of the European Union and the United Kingdom at 250 m resolution. The maps are derived by projecting the global crop-specific rates of PEST-CHEMGRIDS (Maggi et al., 2019) in a 250 m crop map, applying national authorizations and bans, and calibrating the result against pesticide use statistics reported to EUROSTAT. For each of the 55 active ingredients included, a low, median and high estimate is provided in kg/ha. The median estimate was used, and an indicator was obtained by summing the maps of all active ingredients.

This variable should be read as an index of relative pesticide pressure rather than as a complete pesticide load. The authors state that their maps do not include all pesticides authorized in the European Union. Summing across active ingredients also weighs each substance by applied mass alone, without regard to its toxicity to arthropods.

2.3.6. Nitrogen Fertilization

Nitrogen application was derived from NPKGRIDS (Nguyen et al., 2024), a global dataset of nitrogen, phosphorus and potassium application rates for 173 crops with a reference year of 2020 and a resolution of 0.05°, approximately 5.6 km at the equator. Of the three nutrients available, only nitrogen was retained as a proxy for fertilization.

NPKGRIDS reports, for each crop, the nitrogen rate applied per hectare of that crop's harvested area, so the rates cannot be added across crops. Each crop-specific rate was therefore weighted by the harvested area of the same crop taken from CROPGRIDS (Tang et al., 2024), and the total was divided by the land area of the cell. The resulting variable expresses nitrogen applied per hectare of land rather than per hectare of cropped land.

2.3.7. Compositional heterogeneity

Two indicators of the diversity of crops grown in the surrounding landscape were used.

Shannon crop diversity was computed from CROPGRIDS (Tang et al., 2024), which reports the area harvested for each of 173 crops in each cell of a 0.05° grid for 2020. For each cell, the proportion of the total harvested area accounted for by each crop was calculated, and the Shannon index computed on those proportions using the natural logarithm (Ortiz-Burgos, 2016). The index equals zero where a single crop accounts for all of the harvested area of a cell and increases as the harvested area is divided more evenly among crops.

Crop α-diversity was taken from Machefer et al. (2024), who quantify crop diversity from a satellite-derived crop map at 10 m resolution for 2018. For each 1 km cell they compute the Shannon index and exponentiate it. The measure is not a count of crop types but the number of equally abundant crop types that would produce the observed diversity, so a cell dominated by one crop scores close to one while a cell with several crops in similar proportions scores close to the number present. The mean value within each buffer was extracted.

2.3.8. Configurational heterogeneity

Field size predominance was obtained from the global field size dataset of Lesiv et al. (2019), produced through a crowdsourcing campaign in which participants visually interpreted high resolution satellite imagery through the Geo-Wiki application. Fields were defined as enclosed agricultural areas, including annual and perennial crops as well as pastures, hayfields and fallow land, and were assigned to one of five size classes. The product has a resolution of 0.005°, approximately 550 m at the equator.

Each cell records only which field-size class is dominant within it, not the distribution of field sizes it contains and not the size of any individual field. Two variables were derived as the percentage of the buffer occupied by cells where small or very small fields dominate, and by cells where large or very large fields dominate. They are therefore measures of the extent of landscape dominated by a given field-size class, rather than of mean field size.

Other indicators of configurational heterogeneity were computed using the landscapemetrics R package (Hesselbarth et al., 2019), including clumpiness indices for cropland and natural vegetation, but were not retained because the index is undefined for a class absent from a buffer, which produced missing values at a large number of sites.

2.4. Modelling framework and statistical analysis

Total abundance and taxa richness were modelled as response variables in order to study the effect of the local habitat category and of each landscape predictor. Below is the description of the choice of modelling framework, model architecture and statistical analysis.

2.4.1. Choice of modelling framework

The main difficulty of working with the PREDICTS database is described by Purvis et al. (2018), who note that its data "have come from very many source publications, and each sampled different taxonomic groups, using different methods and levels of sampling effort, in sites of different sizes, and in different parts of the world". As a result, biodiversity values from different datasets cannot be compared directly, and the variability caused by sampling methods, taxonomy and geography is unrelated to the effects of interest. The authors describe hierarchically structured statistical models as the way to deal with this problem.

After reviewing Purvis et al. (2018) and the modelling approaches used in previous work based on PREDICTS (Newbold et al., 2015; Fan et al., 2024; Ceaușu et al., 2025; Scheepens et al., 2026), mixed effects models were selected. In this way the variation coming from the structure of the database is absorbed by the random effects, and the fixed effects can be interpreted as the effect of the local habitat and of the surrounding landscape.

2.4.2. Random effects

The random effect's structure was chosen by fitting a series of null models and comparing their AIC values (Appendix 8).

Random effects tested include "Study", to account for different study designs; "Block", to account for the spatial structuring of the sites within each study; "Coordinate ID", to account for time series being present at each physical sampling location; "Sampling method," to account for the ability of the different sampling methods to detect individuals; and "Year", to account for the effect of particularly extreme years.

The retained structure was (1|Study/Block/Coordinate ID) + (1|Sampling Method). The random effect of Year was not included as adding it did not improve the models.

2.4.3. Selection of predictors

A correlation matrix was estimated between all landscape predictors to select the fixed effects of the model. Every pair with an absolute correlation above 0.7 was inspected, retaining only one of the two variables. The full matrix results are presented in Appendix 9. The pairs above the threshold are listed in Table 2-3.

Table 2-3. Pairs of predictors with correlation above 0.7. Underlined, variables retained in the model.

Variable 1	Variable 2	Correlation
Cropland cover	<u>Water cover</u>	−0.83
Annual precipitation	<u>Water cover</u>	0.83
Cropland cover	<u>Natural vegetation cover</u>	−0.82
Shannon crop diversity	<u>Water cover</u>	−0.78
Nitrogen fertilization	<u>Pesticide application</u>	0.75
Annual precipitation	Cropland cover	−0.71

The variable most directly linked to the hypotheses was kept. Natural vegetation cover and water cover were retained over cropland cover and annual precipitation because they are the two variables through which the buffering hypothesis is tested, and pesticide application was retained over nitrogen fertilization as the indicator of agricultural input pressure. Shannon crop diversity was dropped and crop diversity is therefore represented by the alpha diversity indicator.

The predictors retained for both response variables are therefore: local habitat category (10 categories), urban cover, water cover, natural vegetation cover, pasture cover, pesticide application, crop alpha diversity, small fields prevalence, large fields prevalence, and mean temperature.

All continuous predictors were centered and scaled before fitting, so that their coefficients are comparable with one another. Sites with a missing value for any predictor were excluded, so that all models are fitted on the same set of sites.

Interactions were added between the habitat category and all landscape predictors, since we hypothesize that the effect of the landscape composition and agricultural practices on biodiversity may differ depending on the local habitat.

2.4.4. Distributions, model selection and evaluation

Total abundance was modelled with a Gaussian distribution, using the effort-corrected abundance transformed as log(abundance + 1) as the response variable, with the lmer() function of the lme4 package. The transformation was applied because the distribution of the raw values is strongly right-skewed, and the log-transformed response is the form used for abundance in previous work based on PREDICTS (Newbold et al., 2015; Fan et al., 2024; Scheepens et al., 2026). This choice was confirmed through the model fit checks described below.

As a complementary check, models assuming Poisson, negative binomial, zero-inflated negative binomial and Tweedie distributions were fitted on untransformed abundance with the glmmTMB package. They did not improve on the Gaussian model and are reported in Supplementary Information S2. A

Taxa richness was modelled with a zero-inflated negative binomial distribution using the glmmTMB() package, with a nbinom2() family, and a constant zero-inflation term. The need for the zero-inflation component was verified by applying the check_zeroinflation() test from the Dharma package to the equivalent model without it

For each response variable, all subsets and combinations of the predictors were tested and ranked by AICc using the dredge() function of the MuMIn package. Marginality was enforced, so that an interaction could only enter a model together with both of its main effects. Final estimates results are presented here for the average model, considering all models with an AICc difference of less than 2 units to the model with the lowest AICc value.

Model fit was checked with several functions of the performance R package and by inspecting QQ plots of the residuals. For the models fitted with glmmTMB, residuals were additionally examined with the

DHARMa package using 12500 simulations[1]. Model structures, estimates, random effect variances, conditional and marginal R^2, maximum Cook's distances and plots of model fit for all best AIC-models are available in Supplementary Information S2. DHARMa plots of the best AIC-models are shown in Appendix 10.

Natural vegetation was set as the reference level of the habitat category, so that every habitat coefficient reads as the difference relative to the least modified habitat. Since both response variables were modelled on a logarithmic scale, coefficients and their confidence intervals were converted into percentage changes relative to that reference as (exp(estimate) − 1) × 100, and results are presented in those terms.

Finally, because the number of predictors and interactions considered was large, the LASSO was tested as a complementary approach to arriving at a final model. Instead of fitting every combination, it fits a single model containing all predictors and adds a penalty proportional to the size of the coefficients, so that the weakest are shrunk to zero. The details and results are given in Appendix 11.

[1] Although the recommended number of simulations in the package instructions is between 250 and 1000, for large datasets such as the one used here, evidence has been found of the DHARMA package author recommending to use a number of simulations of 5 times the sample size (https://github.com/florianhartig/DHARMa/issues/171)

3. Results

3.1. Response of arthropod abundance and richness to local land use and use intensity.

Based on the best AIC-performing abundance models (AICc = 6360.20, ΔAICc < 2 for 10 models; R^2 conditional= 0.862 ±0.0008, R^2 marginal= 0.046 ± 0.002) and taxa richness models (AICc = 13999.65, ΔAICc < 2 for 10 models; R^2 conditional= 0.954 ±0.0004, R^2 marginal= 0.042 ±0.004), habitats in the more natural, unmanaged group such as natural vegetation and semi-natural vegetation under minimal or light use generally showed higher abundance and richness than the more intensively managed habitat, that is, cropland at all three levels of use intensity and intensively used pasture. As detailed below, this pattern was not uniform across every habitat and intensity level (Figure 3-1).Summary tables with results are also available in Appendix 12 and all models' results details in Supplementary Information S2.

(i) **Cropland.** This habitat showed consistently negative mean estimates relative to natural vegetation, irrespective of intensity level. A significant richness reduction in croplands was observed, but the effect on abundance did not reach significance. For abundance, the decrease was only individually significant in light use intensity with a difference of -34.9% (p= 0.015) compared to natural vegetation. For richness the pattern was more consistent, with all three cropland intensity levels showing significantly lower richness than natural vegetation, and no difference between them. As in abundance, the light intensity croplands produced the highest loss of richness with -26.5% (p= 0.002), while the mean estimate for intense and minimal is similar with -18.9% (p= 0.009) and -17.7% (p= 0.008) respectively.

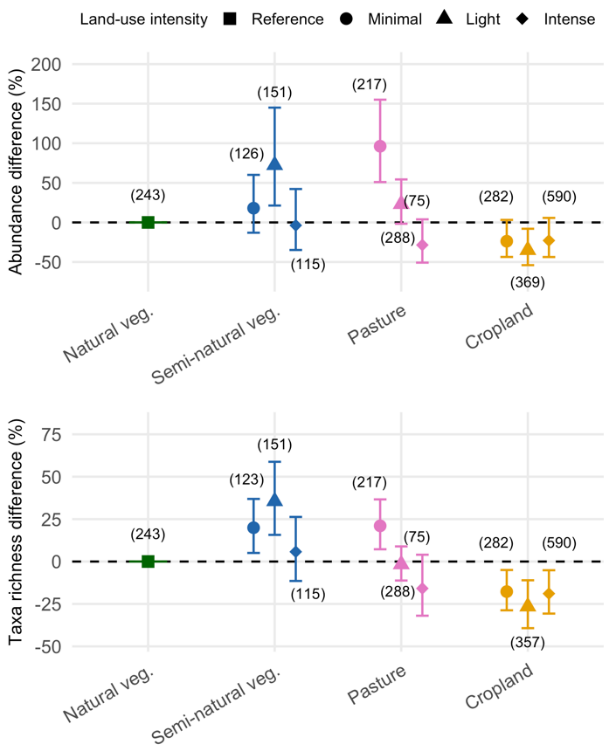


Figure 3-1 Responses by habitat category. Percentage difference in total abundance (up) and taxa richness (down) across habitat categories and land-use intensities, relative to natural vegetation. Points show model average estimates with a 95% confidence interval. Colour indicates land-use type and shape, intensity. Estimates are averaged across all top AIC models (delta AIC <2), which includes 10 models for abundance and 5 for richness. In parenthesis, the number of sites per category.

(ii) **Pasture.** Unlike the other habitats, pasture showed a clearer gradient across use intensity, with intensively used sites holding lower abundance and richness than minimally used ones. Intensively used pasture showed mean estimates that although lower than natural vegetation in both abundance (-28.5%, p= 0.079) and richness (-15.9%, p=0.111) are not significantly different. On the contrary, minimally used pastures presented significantly higher abundance (+96.2%, p<0.001) and richness (21.0%, p=0.002) than natural vegetation, showing that the least disturbed habitat did not have the highest biodiversity response. Lightly-used pastures did not show a significantly different response from that of natural vegetation (abundance: +23.2%, p=0.069; richness: -1.65%, p=0.748).

(iii) **Semi-natural vegetation.** This habitat did not show a decline with increasing use intensity the way pasture did. In fact, semi-natural vegetation with light use showed the highest diversity values in the category holding 72.4% more individuals (p= 0.002) and 35.6% higher taxa richness (p < 0.001) than natural vegetation. Minimally used semi-natural vegetation was only marginally (and not significantly) elevated in abundance (+18.0%, p= 0.288) and significantly higher in richness (+19.9%, p= 0.007) as compared to natural vegetation. Meanwhile, intensely used semi-natural vegetation presented values close to the natural-vegetation baseline in abundance (-3.7%, p= 0.849) and richness (+5.7%, p= 0.537).

3.2. Response of arthropod abundance and richness to landscape composition and practices

Regarding landscape composition and agricultural practice, five out of the nine variables tested (Crop diversity, pesticide use, urban cover, natural habitat cover and small field prevalence) showed either a positive or negative effects on the abundance and/or richness, with some effects depending on local habitat (interactions) (Figure 3-2). The other landscape predictors (pasture cover, mean temperature, water cover, and large field prevalence) showed no significant influence in either model. All results described below are changes <u>per standard deviation increase in the predictor</u> (Table 8-6, Appendix 9).

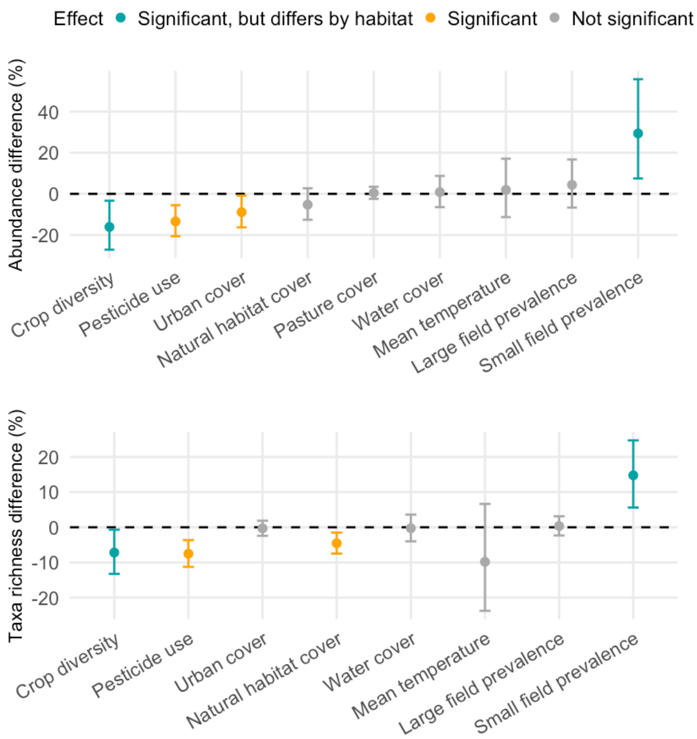


Figure 3-2 Responses by Landscape composition and agricultural practices. Percentage difference in total abundance (up) and taxa richness (down) per unit increase in each landscape composition or agricultural practices. Points show model average estimates with a 95% confidence interval. For predictors that interact significantly with habitat categories in the model, the effect shown here is the slope specifically within Natural vegetation, significant habitat specific slopes are shown in Figure 3-3

Pesticide use in the landscape showed a consistent negative effect, significantly reducing abundance by 12.9% and richness by 7.5% (p = 0.002 and p < 0.001). **Urban cover** significantly reduced abundance by 9.0% (p= 0.028), but had no detectable effect on richness (-0.3%, p= 0.788). In contrast, the percentage of **natural cover** in the landscape was associated with a significant 4.5% reduction in richness (p= 0.004), and no significant change in abundance (-5.5%, p= 0.172).

In none of these three cases there was evidence that the effect differed by habitat, as the corresponding interaction terms were tested but not retained in the best AIC-performing models. By contrast, for both crop diversity and small field prevalence, the habitat interaction was retained in the top-ranked models, which shows that their relationship with abundance and richness depends on the habitat (Figure 3-3)

Crop diversity's showed a negative effect in both abundance (-16.0%, p = 0.016) and richness (-7.2%, p= 0.031) of insects and arachnids in natural vegetation, but in cropland the relationship reverses and is more significant with an increase in use intensity. Greater crop diversity showed a significant positive effect in abundance (+24.4%, p= 0.003) and richness (+16.3%, p < 0.001) in intensively used cropland. This positive effect of increasing crop diversity is also observed in lightly used cropland with significant increase in richness (+14.3%, p= 0.013), and a positive but not significant trend in abundance (+18.0, p=0.058). In minimally used cropland, there was not a significance in either metric, as it was also the case of pasture and most semi-natural vegetation habitats, the exception being minimally used semi-natural vegetation where abundance decreased with crop diversity in the landscape (-15.9%, p= 0.042).

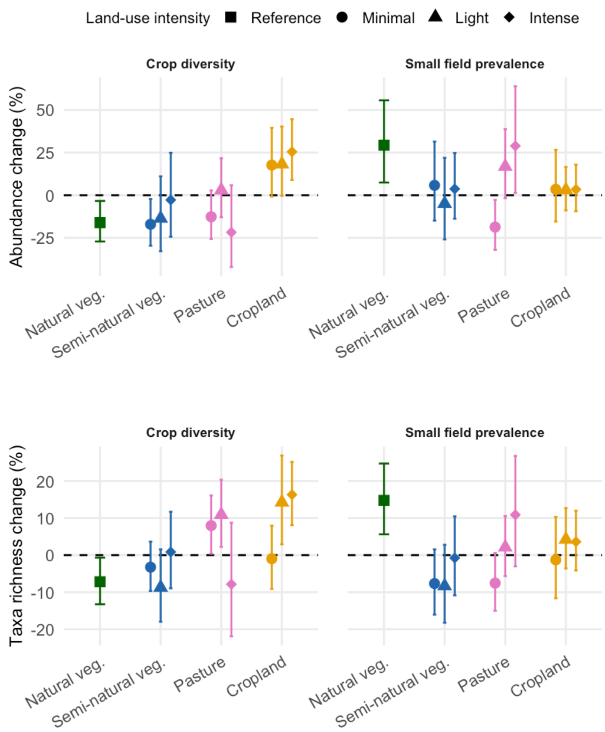


Figure 3-3 Responses in interaction terms. Slope of total abundance (up and taxa richness(down) on crop diversity and small field prevalence. estimated within each habitat category. Only interactions retained in at least one top AIC (delta AIC <2) are shown. When interactions were modelled but are not shown in this Figure it means that there is no evidence to support that the slope of other habitat categories is different for that shown for natural vegetation. Points show model average estimates with a 95% confidence interval. Colour indicates land-use type and shape land-use intensity

Small field prevalence significantly increased both abundance (+29.3%, p= 0.007) and richness (+14.8%, p= 0.001) in natural vegetation, which did not happen in most of the other local habitats. None of the three cropland intensity levels showed a significant increase or decrease in either abundance or richness. Within pasture, the effect on abundance reversed by intensity: intensely used pasture showed a significant increase of abundance with small fields prevalence in the landscape (+29.0%, p = 0.038), while minimally used pasture showed a significant decrease (-19.0%, p = 0.021).

4. Discussion

This analysis produced four main results. First, local habitats under little or no management — natural vegetation and semi-natural vegetation of minimal or light use — hold higher taxa richness of insects and arachnids than intensively managed agricultural habitats, with abundance following the same direction but with weaker support. However, the least disturbed habitats are not the richest as minimally used pasture and lightly used semi-natural vegetation exceed natural vegetation in both metrics. Second, the response of increasing intensity only has a clear gradient in pasture; in cropland the reduction is of similar magnitude in all intensity classes. Third, agricultural practices in the surrounding areas affect arthropods communities with average total pesticide application reducing abundance and richness with no evidence that their effect differs between habitats. Finally, arthropods respond differently to changes in landscape composition and configuration depending on the local use and intensity. Below, each result is further discussed in relation the research hypotheses.

Although significant, the results need to be interpreted within the context of the proportion of the variance explained by the fixed effects in the models. In both models, conditional R^2 were high (0.86 to 0.95) but the marginal R^2 were low (0.034 to 0.051). This means that most of the variance was explained by the random effects (i.e., the structure of the data) rather than by the predictors. The distribution of each significant landscape predictor by study is shown in Appendix 9, where the differences in landscape context between studies are visible. This situation is not particular to this work (Appendix 1). Low marginal R^2 values appear consistently across studies using PREDICTS with comparable modelling approaches, though they are usually reported in supplementary materials rather than discussed in the articles. Similar to the explanation provided by Millard et al. (2021), we consider that the aim of this work is to find trends rather than predict biodiversity for a specific location, so results are still valuable. However, as we argue further in Limitations and perspectives (Section 5), other biodiversity measures and modelling approaches could be used to improve model performance.

The **first hypothesis stating that more natural unmanaged habitat would host higher biodiversity response than less natural and intensively managed habitats** is generally supported by the evidence although with exceptions, and stronger evidence for richness than for abundance. In general, natural habitats and semi-natural habitats of light and minimal use show greater biodiversity than intensive croplands and pastures but in the abundance model, confidence intervals of some of the habitat categories included the reference value, signaling that the result is more nuanced than expected and that the hypothesis hold true in direction of the trend, although not always significant.

Natural vegetation was not the habitat where insects and arachnids presented the highest diversity. Minimally used pasture hosted 96% more individuals than natural vegetation and lightly used semi-natural vegetation 72% more individuals and 36% higher taxa richness. Two elements may contribute, both relating to what the categories contain. First is the categorization of grasslands, which are commonly assigned in PREDICTS to pasture or to secondary vegetation of indeterminate age, the latter was grouped here into semi-natural vegetation. European semi-natural grasslands are not depleted forests but biodiverse systems, surpassing tropical rainforest in vascular plant species richness at small scales and supporting twice as many endemic vascular plants as forests despite covering less land (Shipley et al., 2024). For arthropods specifically, Lacasella et al. (2015) recorded higher species richness of ground-dwelling arthropods in mountain grasslands than in adjacent forest in central Italy, and Duflot et al. (2015) found higher average species richness of carabid beetles in temporary and permanent grasslands than woodland in western France.

Further study of the role of grasslands in driving these results is needed as it has great policy relevance. Grasslands have historically been underrepresented in national and international policy (Bardgett et al., 2021), but this is changing in Europe. The EU Nature Restoration and Amending Regulation, in force since August 2024, creates the obligation for Member States to increase the trend of at least two out of three indicators of agricultural ecosystems to demonstrate restoration, one being the Grassland

Butterfly Index (Regulation (EU) 2024/1991, Article 11). That index fell by 47% for 17 grassland butterfly species between 1991 and 2024 (European Environment Agency, 2025), so understanding what drives arthropod abundance and richness in grassland is of extreme importance.

The second element is the composition of the natural vegetation category itself, which combines primary vegetation and mature secondary vegetation. In Europe, forests with minimal signs of human use represent less than 4% of the forest area, the primary forests that remain are scarce and highly fragmented, and it is unclear whether they are representative of the range of natural forest types found in the region (Sabatini et al., 2020). Scheepens et al. (2026) reach a comparable conclusion using PREDICTS, attributing the comparatively weak agricultural effects they observe in Europe partly to a degraded primary vegetation baseline.

The second hypothesis predict that abundance and richness would fall more steeply with increasing management intensity in less natural habitats than in the more natural ones is only partly supported. A clear intensity gradient appears only in pasture, where intensively used sites hold lower abundance and richness than minimally used ones. In cropland, the response is essentially flat (i.e., there is an absence of a gradient) similarly to what was observed by Scheepens et al. (2026), who, working with an un-curated data of all arthropods, found abundance and richness in European croplands to be reduced relative to primary vegetation but independent of use intensity. Meanwhile, in semi-natural vegetation, biodiversity peaks at sites with light use, which is consistent with the intermediate disturbance hypothesis that proposes that diversity peaks at intermediate levels of disturbance. In this hypothesis, where disturbance is low, a few dominant competitors gradually exclude weaker ones, while where it is high only tolerant, fast-colonizing species persist (Connell, 1978). Although refuted by some authors, notably Fox (2013), the theory could be here explicatory of the situation.

Reasons behind the lack of gradient in most cases may also lie in how intensity classes are constructed. Use intensity in PREDICTS is not measured but inferred by the compilers from the description given in the original publication that is most commonly not quantitative (Hudson et al., 2014; Hudson et al., 2017;). Where those descriptions are brief or imprecise, the resulting classes may distinguish poorly different levels of management, which would attenuate any real gradient. Additionally, use intensities are commonly assigned at the study level, so that intense use in one study may not be comparable to intense use in another. Another reason could be that intensification acts as an ecological filter, replacing specialists with generalists adapted to certain types of disturbance (Gámez-Virués et al., 2015), a change in composition (turnover) that abundance and richness cannot capture.

The third hypothesis stating that intensively managed landscape would host lower abundance and richness than less managed landscape is supported for pesticides, could not be tested for fertilization, and is only partly supported for heterogeneity. Pesticide application in the surrounding landscape reduced abundance and diversirty and the effect did not differ between habitats To our knowledge this is the first analysis to relate total pesticide application rates to the abundance and richness of Insecta and Arachnida across various habitat types at the European regional scale. The closest precedent is Millard et al. (2021), who using PESTCHEMGRIDS and PREDICTS built models to understand invertebrates' response to pesticides globally for croplands habitats only, finding a similar negative effect in total abundance (except for Diptera) and a richness response highly dependent on the Order, with negative effect on Lepidoptera but positive in Diptera and Coleoptera.

Fertilization could not be assessed independently. Nitrogen application and pesticide application were correlated at 0.75 in the predictor set, and only one of the two could be retained (Section 2.4). Millard et al. (2021) encountered the same limitation: running separate models for each variable, they found the response of flies to pesticide application rate to be similar to that of fertilization. They described the cause as unclear, hypothesizing that "it may be that fertilizer and pesticide application rate are correlated", which is indeed what was observed here across a broader range of habitats.

For compositional and configurational heterogeneity, the direction of the effect depends on the local habitat and indicator. In intensively used croplands, having greater crop diversity in the surrounding landscape was associated with higher abundance (+24.4%) and richness (+16.3%) of arthropods (as stated in the hypothesis), but the opposite happens when the local habitat is natural vegetation. In those cases, increasing crop diversity was associated with lower abundance (−16.0%) and richness (−7.2%). This reversal is the part we did not expect, and two possible explanations were examined. The first is that the indicator could be describing the extent of cropland rather than its diversity, since the α-diversity layer assigns a value of zero to cells without cropland and the mean estimated for cell within the buffer therefore mixes the two. To check this, the variable (crop diversity) was recomputed using only the cells that contain cropland. The two versions correlate at 0.94, so the indicator is mostly capturing how diverse the cropping is and not how much cropland there is. The second is there can be a compounding effect between crop diversity and the percentage of natural vegetation within the landscape. When estimating the correlation matrix to identify collinearity in the process of defining the model, all sites in the dataset were used, however, when inspected among natural vegetation sites alone, crop α-diversity and the percentage of natural vegetation in the buffer correlate at −0.63, so the sites surrounded by more diverse cropping are also those with less natural cover around them. This explanation remains open and would need further analysis. Finally, small field prevalence behaved in the opposite way, increasing abundance and richness in natural vegetation, as it was expected but showing no detectable effect in any cropland category.

The fourth hypothesis, that the changes described above would be more pronounced in more natural local habitats than in already degraded intensively used ones, was not supported. The interactions were not retained for the land cover indicators or for pesticide application, so there is no evidence that their effect differs between habitats, and where they were retained, for crop diversity and small field prevalence, the effects were not distinguishable from zero across use intensity levels in the less disturbed habitats, so no comparison could be drawn. This differs from what has been found for other pressures. Fan et al. (2025) were able to identify the strongest effects of nitrogen deposition where local land use was least modified. Two things may explain the difference. The least modified end of the gradient in this dataset does not include genuinely intact primary vegetation, since European forests with minimal signs of human use cover less than 4% of the forest area (Sabatini et al., 2020), and the range of the landscape predictors within each habitat is narrow, which makes it hard to detect the differences in slope.

Finally, contrary to what was expected, the evidence did not support the fifth hypothesis, which stated that more natural vegetation in the surrounding landscape would buffer the decline in abundance and richness caused by pressures. While no significant effect was found for the abundance of insects and arachnids, natural cover in the landscape was associated with a significant 4.5% reduction in richness ($p = 0.004$). This differs from Outhwaite et al. (2022), who found that more nearby natural habitats can mitigate the reductions in insect abundance and richness, and from Fan et al. (2025), who found that semi-natural and natural habitats in the surrounding landscape reduce arthropod losses associated with nitrogen deposition. One possible explanation is that a higher proportion of any single cover in the landscape (e.g., natural vegetation cover) could describe a more homogeneous landscape, and that in that case, a negative association with richness reflects more a reduced landscape heterogeneity than an effect of natural vegetation as such. Further analysis should be needed to understand whether this mechanism, which has been described in literature to vary between species groups (Tews et al., 2004) can be in fact driving the results.

Taken together, these results show that the local habitat, the agricultural practices of the surrounding landscape, and the landscape land covers and composition shape arthropod abundance and richness in Europe, but that the direction and detectability of those effects depend heavily on how habitats and practices are represented in the underlying data. The limitations this imposes on the present analysis are considered below.

5. Limitations and perspectives

The limitations of this analysis are presented below, together with the perspective it opens for further work.

- **Resolution and temporal alignment of the agronomic data.** The landscape variables come from products of very different resolutions, from 30 m for cropland cover to 0.05°, roughly 5.6 km, for nitrogen application. Where cells are coarse, the value assigned to a 1 km buffer rests on one to four grid cells, so the variable describes the wider agricultural context rather than the immediate surroundings of the site. The reference years are also sometimes not well aligned with the biodiversity data. Sampling in the curated dataset ends in 2011, whereas the pesticide maps describe 2015 to 2020, and the fertilizer and crop diversity layers describe 2018 and 2020 (Table 2-2). The landscape was therefore, in some cases, characterized several years after the arthropods were sampled. Although we assume that agricultural patterns are broadly stable over that interval, in further work it would be advisable to implement mechanisms to address the issue.
- **Taxonomic resolution.** Insects and arachnids were analysed together, and taxonomic resolution varies across the dataset from species to class. Evidence that responses differ between orders (Millard et al., 2021) suggests that further analyses by taxonomic or functional group would be informative.
- **The pesticide indicator.** Active ingredients were summed by applied mass, which weights highly toxic and low-toxicity substances equally, and the source maps do not cover all substances authorized in the European Union. Since the toxicity of applied insecticides has risen while the applied quantity (mass) has fallen(Schulz et al., 2021), indicators weighing application quantity by toxicity would represent the pressure on arthropods more precisely than quantity alone does.
- **Indicators of heterogeneity.** Compositional and configurational heterogeneity were each represented by a single indicator in the model, and both carry simplifications. The field size layer records only which size class is dominant in a cell, and crop diversity is a mean across the buffer that combines the diversity of cropping with its extent. Alternative measures would help test whether the responses dependent on the habitat we found here are robust.
- **Practices not represented.** No spatially explicit information was available for other practices such as local organic management, tillage, or crop rotation. These omitted practices could be correlated with those that were considered, and so they could be enhancing or diminishing the observed effects.
- **Spatial scale.** All landscape variables were extracted within a single 1 km buffer. Landscape effects on arthropods operate at different distances depending on the group and the process, so comparing several radii would help establish whether the results are highly dependent on this design parameter.
- **Biases in the source database.** PREDICTS carries spatial and taxonomic biases, and its European records are relatively old (before 2011). Integrating additional biodiversity databases would improve coverage, but would require an alternative way of defining local land use and use intensity, since these are recorded in a form specific to PREDICTS. A sensitivity analysis comparing subsets of the data would allow to assess the influence of these biases.
- **Model performance.** Marginal R^2 is low, reflecting the heterogeneity of the source studies rather than the absence of an effect. The range of the landscape predictors also differs considerably between habitats, which limits the power to detect differences in slope. Other model structures could be tested, in particular formulations that compare land uses and use intensities directly against the least disturbed category within each study, so that results are reported as within-study contrasts.

- **Biodiversity database issues.** The curation process found issues in most of the sources reviewed, and only some could be corrected. In many instances we chose only to report the differences, as we did not have the information needed to determine whether they resulted from explicit and informed decisions taken when the data were compiled. The values these sources contribute therefore carry a known and unquantified uncertainty. The full list of issues and the action taken for each is given in Supplementary Information S1.

Separating the effects of individual agricultural practices on arthropods at large scales remains highly complex, with three factors standing out. The spatial data describing what is actually applied in the field remain coarse or absent. The biodiversity databases available carry spatial and taxonomic biases and uneven temporal coverage. Modelling is challenging because the differences between studies and sampling designs are large. As a result, the predictors of interest explain only a small share of the variation in the response, even though the models reproduce the data well.

6. Conclusions

This study evaluated how agricultural practices and landscape composition affect the abundance and richness of insects and arachnids across habitat types in Europe. Habitats under little or no management held more taxa than intensively managed ones, as expected, but the least disturbed habitat was not the richest: minimally used pasture and lightly used semi-natural vegetation held more arthropods than natural vegetation. Protecting only the habitats that appear least modified is therefore not enough. In a region where undisturbed forest is scarce, lightly managed habitats such as pastures and semi-natural vegetation, which in this dataset often correspond to grasslands, hold a large share of the arthropod diversity and deserve as much conservation attention. Pesticide application in the surrounding landscape reduced both abundance and richness in every habitat, while crop diversity and field size mattered but changed direction depending on the local habitat. What is applied around a site therefore reaches beyond farmland, so protecting a habitat without considering how the land around is managed may not be enough. And measures promoted for farmland biodiversity, such as diversifying crops or reducing field size, cannot be assumed to have the same effect in every habitat.

In this work we were also interested in understanding if global biodiversity databases and agronomic datasets could, as sources, be used to answer those questions in a way that allows for the disentangling of the effects of particular practices or surrounding conditions in agricultural landscapes. In fact, the database supports comparisons between habitat types and it was possible to detect effects, especially of pesticides at landscape scale. However, it was not possible to separate pesticides from fertilization (due to correlation), to include other practices (due to lack of spatial products), to resolve differences between intensity levels within croplands (due, potentially, to use intensity definitions in the database), or to explain a higher portion of the variance through the fixed effects (due to data structure or modelling approach). Given all this, we would encourage further work at the European or global scale to test alternative modelling approaches, maintaining the curation processes of the biodiversity database we applied here, as relevant issues were discovered in almost all references included.

Finally, we conclude that this work contributes to two main aspects. First, it is, to our knowledge, the first analysis relating pesticide application to the abundance and richness of insects and arachnids across habitat types at European scale. Second, it produces a curated dataset, checked reference by reference against the original publications, which can be used in further work. Checking a compiled database such as PREDICTS is slow, but given what we found here, and the issues other authors have already pointed it, it is essential. Understanding which individual practices drive biodiversity changes, and by how much, remains an open question at large scale. It matters because international assessments and policy targets are built on evidence about drivers, and evidence that treats agriculture as a single pressure cannot tell decision-makers which practices acting on.

7. Bibliographic References

[1]. Alkemade, R., Van Bussel, L. G., Rodríguez, S. L., & Schipper, A. M. (2022). Global biodiversity assessments need to consider mixed multifunctional land-use systems. *Current Opinion in Environmental Sustainability*, *56*, 101174. https://doi.org/10.1016/j.cosust.2022.101174

[2]. Bardgett, R. D., Bullock, J. M., Lavorel, S., Manning, P., Schaffner, U., Ostle, N., Chomel, M., Durigan, G., L. Fry, E., Johnson, D., Lavallee, J. M., Le Provost, G., Luo, S., Png, K., Sankaran, M., Hou, X., Zhou, H., Ma, L., Ren, W., … Shi, H. (2021). Combatting global grassland degradation. *Nature Reviews Earth & Environment, 2(10),* 720–735. https://doi.org/10.1038/s43017-021-00207-2

[3]. Bartoń, K. (2026). *MuMIn: Multi-model inference* (R package version 1.48.19) [Computer software]. https://doi.org/10.32614/CRAN.package.MuMIn

[4]. Batáry, P., Báldi, A., Kleijn, D., & Tscharntke, T. (2011). Landscape-moderated biodiversity effects of agri-environmental management: A meta-analysis. *Proceedings of the Royal Society B: Biological Sciences*, *278*(1713), 1894–1902. https://doi.org/10.1098/rspb.2010.1923

[5]. Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software, 67*(1), 1–48. https://doi.org/10.18637/jss.v067.i01

[6]. Brooks, M. E., Kristensen, K., van Benthem, K. J., Magnusson, A., Berg, C. W., Nielsen, A., Skaug, H. J., Maechler, M., & Bolker, B. M. (2017). glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *The R Journal, 9*(2), 378–400. https://doi.org/10.32614/RJ-2017-066

[7]. Ceaușu, S., Leclère, D., & Newbold, T. (2025). Geography and availability of natural habitat determine whether cropland intensification or expansion is more detrimental to biodiversity. *Nature Ecology & Evolution*, *9*(6), 993–1008. https://doi.org/10.1038/s41559-025-02691-x

[8]. Ceballos, G., Ehrlich, P. R., & Dirzo, R. (2017). Biological annihilation via the ongoing sixth mass extinction signaled by vertebrate population losses and declines. *Proceedings of the National Academy of Sciences*, *114*(30). https://doi.org/10.1073/pnas.1704949114

[9]. Copernicus Land Monitoring Service. (n.d.). User Manual Copernicus Lands Monitoring Service High Resolution land cover characteristics Lot4: Water & Wetness 2018. https://land.copernicus.eu/en/technical-library/hrl-water-wetness-2018-user-manual (Accessed on 18.05.2026).

[10]. Daskalova, G. N., Phillimore, A. B., & Myers-Smith, I. H. (2021). Accounting for year effects and sampling error in temporal analyses of invertebrate population and biodiversity change: A comment on Seibold et al . 2019. Insect Conservation and Diversity, 14(1), 149–154. https://doi.org/10.1111/icad.12468

[11]. d'Andrimont, R., Verhegghen, A., Lemoine, G., Kempeneers, P., Meroni, M., & Van Der Velde, M. (2021). From parcel to continental scale – A first European crop type map based on Sentinel-1 and LUCAS Copernicus in-situ observations. *Remote Sensing of Environment*, *266*, 112708. https://doi.org/10.1016/j.rse.2021.112708

[12]. Desquilbet, M., Gaume, L., Grippa, M., Céréghino, R., Humbert, J.-F., Bonmatin, J.-M., Cornillon, P.-A., Maes, D., Van Dyck, H., & Goulson, D. (2020). Comment on "Meta-analysis reveals declines in terrestrial but increases in freshwater insect abundances." *Science*, *370*(6523), eabd8947. https://doi.org/10.1126/science.abd8947

[13]. Dirzo, R., Young, H. S., Galetti, M., Ceballos, G., Isaac, N. J. B., & Collen, B. (2014). Defaunation in the Anthropocene. *Science*, *345*(6195), 401–406. https://doi.org/10.1126/science.1251817

[14]. Dornelas, M., Antão, L. H., Bates, A. E., Brambilla, V., Chase, J. M., Chow, C. F. Y., Fontrodona-Eslava, A., Magurran, A. E., Martins, I. S., Moyes, F., Sagouis, A., Adu-Acheampong, S., Acquah-Lamptey, D., Adam, D., Ajani, P. A., Albaina, A., Almaraz, P., An, J., Anderson, R. S., … Zvuloni, A. (2025). BioTIME 2.0: Expanding and Improving a Database of Biodiversity Time Series. *Global Ecology and Biogeography*, *34*(5), e70003. https://doi.org/10.1111/geb.70003

[15]. Dullinger, I., Essl, F., Moser, D., Erb, K., Haberl, H., & Dullinger, S. (2021). Biodiversity models need to represent land-use intensity more comprehensively. *Global Ecology and Biogeography*, *30*(5), 924–932. https://doi.org/10.1111/geb.13289

[16]. Dutt, S., Remmel, T. K., Rivas, C. A., Mazziotta, A., & Kunz, M. (2026). Advancing forest fragmentation analysis: A systematic review of evolving spatial metrics, software platforms, and remote sensing innovations. *Landscape Ecology*, *41*(6), 92. https://doi.org/10.1007/s10980-026-02354-7

[17]. Duflot, R., Aviron, S., Ernoult, A., Fahrig, L., & Burel, F. (2015). Reconsidering the role of 'semi-natural habitat' in agricultural landscape biodiversity: A case study. *Ecological Research*, *30*(1), 75–83. https://doi.org/10.1007/s11284-014-1211-9

[18]. European Commission (2020). *EU Biodiversity Strategy for 2030: Bringing nature back into our lives*. COM(2020) 380 final.

[19]. European Environmental Agency (2025). *Grassland butterfly index in Europe*. https://www.eea.europa.eu/en/analysis/indicators/grassland-butterfly-index-in-europe-1 (Acessed on 13.08.2026).

[20]. European Union's Copernicus Land Monitoring Service information (2020). Water and Wetness status 2018 https://land.copernicus.eu/en/products/high-resolution-layer-water-and-wetness/water-and-wetness-status-2018 (Accessed on 18.05.2026). https://doi.org/10.2909/f6bbd22c-52e5-4e47-9b09-943415fcb52e

[21]. European Union's Copernicus Land Monitoring Service information (2020b). CORINE Land Cover 2000 (vector/raster 100 m), Europe, 6-yearly. https://land.copernicus.eu/en/products/corine-land-cover/clc-2000 (Accessed on 09.06.2026). https://doi.org/10.2909/ddacbd5e-068f-4e52-a596-d606e8de7f40

[22]. European Union's Copernicus Land Monitoring Service information (2020c). CORINE Land Cover 2006 (vector/raster 100 m), Europe, 6-yearly. https://land.copernicus.eu/en/products/corine-land-cover/clc-2006 (Accessed on 18.05.2026).https://doi.org/10.2909/08560441-2fd5-4eb9-bf4c-9ef16725726a

[23]. Fan, S., Newbold, T., Axmacher, J. C., Outhwaite, C. L., Zou, Y., Yu, Z., & Liu, Y. (2025). Arthropod biodiversity loss from nitrogen deposition is buffered by natural and semi-natural habitats. *PLOS Biology*, *23*(7), e3003285. https://doi.org/10.1371/journal.pbio.3003285

[24]. Fan, S., Newbold, T., Tscharntke, T., Tang, W., Yu, Z., & Liu, Y. (2024). Impact of Crop Type on Biodiversity Globally. *Global Change Biology*, *30*(12), e70005. https://doi.org/10.1111/gcb.70005

[25]. Fox, J. W. (2013). The intermediate disturbance hypothesis should be abandoned. *Trends in Ecology & Evolution*, *28*(2), 86–92. https://doi.org/10.1016/j.tree.2012.08.014

[26]. Gaume, L., & Desquilbet, M. (2024). InsectChange: Comment. *Peer Community Journal*, *4*, e97. https://doi.org/10.24072/pcjournal.469

[27]. GBIF. (2026). *Global Biodiversity Information Facility*. Retrieved July 28, 2026, from https://www.gbif.org/

[28]. Goulson, D. (2019). The insect apocalypse, and why it matters. *Current Biology*, *29*(19), R967–R971. https://doi.org/10.1016/j.cub.2019.06.069

[29]. Groll, A., & Tutz, G. (2014). Variable selection for generalized linear mixed models by L 1-penalized estimation. *Statistics and Computing*, *24*(2), 137–154. https://doi.org/10.1007/s11222-012-9359-z

[30]. Groll, A. (2026). *glmmLasso: Variable selection for generalized linear mixed models by L1-penalized estimation* (R package version 1.6.4) [Computer software]. https://doi.org/10.32614/CRAN.package.glmmLasso

[31]. Halsch, C. A., Elphick, C. S., Bahlai, C. A., Forister, M. L., Wagner, D. L., Ware, J. L., & Grames, E. M. (2025). Meta-synthesis reveals interconnections among apparent drivers of insect biodiversity loss. *BioScience*, *75*(6), 448–456. https://doi.org/10.1093/biosci/biaf034

[32]. Hartig, F. (2026). *DHARMa: Residual diagnostics for hierarchical (multi-level / mixed) regression models* (R package version 0.5.0) [Computer software]. https://doi.org/10.32614/CRAN.package.DHARMa

[33]. Harvey, J. A., Heinen, R., Armbrecht, I., Basset, Y., Baxter-Gilbert, J. H., Bezemer, T. M., Böhm, M., Bommarco, R., Borges, P. A. V., Cardoso, P., Clausnitzer, V., Cornelisse, T., Crone, E. E., Dicke, M., Dijkstra, K.-D. B., Dyer, L., Ellers, J., Fartmann, T., Forister, M. L., … De Kroon, H. (2020). International scientists formulate a roadmap for insect conservation and recovery. *Nature Ecology & Evolution*, *4*(2), 174–176. https://doi.org/10.1038/s41559-019-1079-8

[34]. Hesselbarth, M. H. K., Sciaini, M., With, K. A., Wiegand, K., & Nowosad, J. (2019). *landscapemetrics*: An open-source *R* tool to calculate landscape metrics. *Ecography*, *42*(10), 1648–1657. https://doi.org/10.1111/ecog.04617

[35]. Hijmans, R. (2026). *terra: Spatial data analysis* (R package version 1.8-93) [Computer software]. https://doi.org/10.32614/CRAN.package.terra

[36]. Hudson, L. N., Newbold, T., Contu, S., Hill, S. L. L., Lysenko, I., De Palma, A., Phillips, H. R. P., Senior, R. A., Bennett, D. J., Booth, H., Choimes, A., Correia, D. L. P., Day, J., Echeverría-Londoño, S., Garon, M., Harrison, M. L. K., Ingram, D. J., Jung, M., Kemp, V., … Purvis, A. (2014). The PREDICTS database: A global database of how local terrestrial biodiversity responds to human impacts. *Ecology and Evolution*, *4*(24), 4701–4735. https://doi.org/10.1002/ece3.1303

[37]. Hudson, L. N., Newbold, T., Contu, S., Hill, S. L. L., Lysenko, I., De Palma, A., Phillips, H. R. P., Alhusseini, T. I., Bedford, F. E., Bennett, D. J., Booth, H., Burton, V. J., Chng, C. W. T., Choimes, A., Correia, D. L. P., Day, J., Echeverría-Londoño, S., Emerson, S. R., Gao, D., … Purvis, A. (2017). The database of the PREDICTS (Projecting Responses of Ecological Diversity In Changing Terrestrial Systems) project. *Ecology and Evolution*, *7*(1), 145–188. https://doi.org/10.1002/ece3.2579

[38]. Hurtt, G. C., Chini, L. P., Frolking, S., Betts, R. A., Feddema, J., Fischer, G., Fisk, J. P., Hibbard, K., Houghton, R. A., Janetos, A., Jones, C. D., Kindermann, G., Kinoshita, T., Klein Goldewijk, K., Riahi, K., Shevliakova, E., Smith, S., Stehfest, E., Thomson, A., … Wang, Y. P. (2011). Harmonization of land-use scenarios for the period 1500–2100: 600 years of global gridded annual land-use transitions, wood harvest, and resulting secondary lands. Climatic Change, 109(1–2), 117–161. https://doi.org/10.1007/s10584-011-0153-2

[39]. Kleijn, D., Baquero, R. A., Clough, Y., Díaz, M., De Esteban, J., Fernández, F., Gabriel, D., Herzog, F., Holzschuh, A., Jöhl, R., Knop, E., Kruess, A., Marshall, E. J. P., Steffan-Dewenter, I., Tscharntke, T., Verhulst, J., West, T. M., & Yela, J. L. (2006). Mixed biodiversity benefits of agri-environment schemes in five European countries. *Ecology Letters*, *9*(3), 243–254. https://doi.org/10.1111/j.1461-0248.2005.00869.x

[40]. Kissling, W. D., & Lumbierres, M. (2023). *Essential Biodiversity Variable workflows: Designing the freshwater, marine and terrestrial EBV workflows from data collection to modeling*. Research Ideas and Outcomes. https://doi.org/10.3897/arphapreprints.e101949

[41]. Kissling, W. D., Lumbierres, M., Lyche Solheim, A., Liquete, C., Breeze, T. D., Bonn, A., McCallum, I., Maes, J., Hirsch, T., Van Grunsven, R. H. A., Beja, P., Smets, B., Capinha, C., Ceia-Hasse, A., Fernández, N., Santana, J., Moreira, F., Junker, J., Leese, F., … Pereira, H. M. (2026). Building the backbone for Europe's biodiversity monitoring. *Nature Reviews Biodiversity*. https://doi.org/10.1038/s44358-026-00140-6

[42]. Lacasella, F., Gratton, C., De Felici, S., Isaia, M., Zapparoli, M., Marta, S., & Sbordoni, V. (2015). Asymmetrical responses of forest and "beyond edge" arthropod communities across a forest–grassland ecotone. *Biodiversity and Conservation*, *24*(3), 447–465. https://doi.org/10.1007/s10531-014-0825-0

[43]. Lassaletta, L., Billen, G., Grizzetti, B., Anglade, J., & Garnier, J. (2014). 50 year trends in nitrogen use efficiency of world cropping systems: The relationship between yield and nitrogen input to cropland. *Environmental Research Letters*, *9*(10), 105011. https://doi.org/10.1088/1748-9326/9/10/105011

[44]. Lesiv, M., Laso Bayas, J. C., See, L., Duerauer, M., Dahlia, D., Durando, N., Hazarika, R., Kumar Sahariah, P., Vakolyuk, M., Blyshchyk, V., Bilous, A., Perez-Hoyos, A., Gengler, S., Prestele, R., Bilous, S., Akhtar, I. U. H., Singha, K., Choudhury, S. B., Chetri, T., … Fritz, S. (2019). Estimating the global distribution of field size using crowdsourcing. *Global Change Biology*, *25*(1), 174–186. https://doi.org/10.1111/gcb.14492

[45]. Liao, Y., Chen, S., Bai, Y., Wang, J., & Gong, P. (2026). *Global 30-m annual cropland extent dynamics (2000–2024): A consistent baseline of structural evolution and regional disparities*. ESSD – Anthroposphere/Land cover and land use. https://doi.org/10.5194/essd-2025-838

[46]. Machefer, M., Zampieri, M., Van Der Velde, M., Dentener, F., Claverie, M., & d'Andrimont, R. (2024). Earth Observation based multi-scale analysis of crop diversity in the European Union: First insights for agro-environmental policies. *Agriculture, Ecosystems & Environment*, *374*, 109143. https://doi.org/10.1016/j.agee.2024.109143

[47]. Maggi, F., Tang, F. H. M., La Cecilia, D., & McBratney, A. (2019). PEST-CHEMGRIDS, global gridded maps of the top 20 crop-specific pesticide application rates from 2015 to 2025. Scientific Data, 6(1), 170. https://doi.org/10.1038/s41597-019-0169-4

[48]. Millard, J., Outhwaite, C. L., Kinnersley, R., Freeman, R., Gregory, R. D., Adedoja, O., Gavini, S., Kioko, E., Kuhlmann, M., Ollerton, J., Ren, Z.-X., & Newbold, T. (2021). Global effects of land-use intensity on local pollinator biodiversity. *Nature Communications*, *12*(1), 2902. https://doi.org/10.1038/s41467-021-23228-3

[49]. Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D. G., Piles, M., Rodríguez-Fernández, N. J., Zsoter, E., Buontempo, C., & Thépaut, J.-N. (2021). ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, *13*(9), 4349–4383. https://doi.org/10.5194/essd-13-4349-2021

[50]. Newbold, T., Hudson, L. N., Hill, S. L. L., Contu, S., Lysenko, I., Senior, R. A., Börger, L., Bennett, D. J., Choimes, A., Collen, B., Day, J., De Palma, A., Díaz, S., Echeverria-Londoño, S., Edgar, M. J., Feldman, A., Garon, M., Harrison, M. L. K., Alhusseini, T., … Purvis, A. (2015). Global effects of land use on local terrestrial biodiversity. *Nature*, *520*(7545), 45–50. https://doi.org/10.1038/nature14324

[51]. Newbold, T., Bentley, L. F., Hill, S. L. L., Edgar, M. J., Horton, M., Su, G., Şekercioğlu, Ç. H., Collen, B., & Purvis, A. (2020). Global effects of land use on biodiversity differ among functional groups. *Functional Ecology*, *34*(3), 684–693. https://doi.org/10.1111/1365-2435.13500

[52]. Nguyen, T. H., Tang, F. H. M., Conchedda, G., Casse, L., Obli-Laryea, G., Tubiello, F. N., & Maggi, F. (2024). NPKGRIDS: A global georeferenced dataset of N, P2O5, and K2O fertilizer application rates for 173 crops. *Scientific Data*, *11*(1), 1179. https://doi.org/10.1038/s41597-024-04030-4

[53]. Ortiz-Burgos, S. (2016). Shannon-Weaver Diversity Index. In M. J. Kennish (Ed.), *Encyclopedia of Estuaries* (pp. 572–573). Springer Netherlands. https://doi.org/10.1007/978-94-017-8801-4_233

[54]. Outhwaite, C. L., McCann, P., & Newbold, T. (2022a). Agriculture and climate change are reshaping insect biodiversity worldwide. *Nature*, *605*(7908), 97–102. https://doi.org/10.1038/s41586-022-04644-x

[55]. Outhwaite, C. L., Ortiz, A. M. D., Spooner, F. E. B., Dalin, C., & Newbold, T. (2022b). Availability and proximity of natural habitat influence cropland biodiversity in forest biomes globally. *Global Ecology and Biogeography*, *31*(8), 1589–1602. https://doi.org/10.1111/geb.13532

[56]. Pebesma, E. (2018). Simple features for R: Standardized support for spatial vector data. *The R Journal, 10*(1), 439–446. https://doi.org/10.32614/RJ-2018-009

[57]. Porta, G. M., Casse, L., Manzoni, A., Riva, M., Maggi, F., & Guadagnini, A. (2025). Pesticides application rate maps in the European Union at a 250 m spatial resolution. *Scientific Data*, *12*(1), 725. https://doi.org/10.1038/s41597-025-05031-7

[58]. R Core Team. (2025). *R: A language and environment for statistical computing* (Version 4.5.2) [Computer software]. R Foundation for Statistical Computing. https://www.R-project.org/

[59]. Sabatini, F. M., Keeton, W. S., Lindner, M., Svoboda, M., Verkerk, P. J., Bauhus, J., Bruelheide, H., Burrascano, S., Debaive, N., Duarte, I., Garbarino, M., Grigoriadis, N., Lombardi, F., Mikoláš, M., Meyer, P., Motta, R., Mozgeris, G., Nunes, L., Ódor, P., … Kuemmerle, T. (2020). Protection gaps and restoration opportunities for primary forests in Europe. *Diversity and Distributions*, *26*(12), 1646–1662. https://doi.org/10.1111/ddi.13158

[60]. Sánchez-Bayo, F., & Wyckhuys, K. A. G. (2019). Worldwide decline of the entomofauna: A review of its drivers. Biological Conservation, 232, 8–27. https://doi.org/10.1016/j.biocon.2019.01.020

[61]. Santana, J., Porto, M., Morán-Ordóñez, A., Brotons, L., Junker, J., Kissling, W. D., Lumbierres, M., Moe, S. J., Solheim, A. L., Villero, D., Breeze, T. D., Carvalho, L., Fernández, N., Herrando, S., López-Baucells, A., Shamoun-Baranes, J., Silero, N., Pereira, H. M., Moreira, F., & Beja, P. (2025). Large Biodiversity Monitoring Gaps Remain Across Europe. *Conservation Letters*, *18*(5), e13134. https://doi.org/10.1111/conl.13134

[62]. Scheepens, D., Cornford, R., Visconti, P., & Newbold, T. (2026). Large Regional Variation in Global Impacts of Agriculture on Terrestrial Insects and Other Arthropods. Global Change Biology Communications, 1(3), e70040. https://doi.org/10.1002/gcb4.70040

[63]. Schelldorfer, J., Bühlmann, P., & De Geer, S. V. (2011). Estimation for High-Dimensional Linear Mixed-Effects Models Using ℓ_1 -Penalization. *Scandinavian Journal of Statistics*, *38*(2), 197–214. https://doi.org/10.1111/j.1467-9469.2011.00740.x

[64]. Schulz, R., Bub, S., Petschick, L. L., Stehle, S., & Wolfram, J. (2021). Applied pesticide toxicity shifts toward plants and invertebrates, even in GM crops. Science, 372(6537), 81–84. https://doi.org/10.1126/science.abe1148

[65]. Shipley, J. R., Frei, E. R., Bergamini, A., Boch, S., Schulz, T., Ginzler, C., Barandun, M., Bebi, P., Bolliger, J., Bollmann, K., Delpouve, N., Gossner, M. M., Graham, C., Krumm, F., Marty, M., Pichon, N., Rigling, A., & Rixen, C. (2024). Agricultural practices and biodiversity: Conservation policies for semi-natural grasslands in Europe. *Current Biology*, *34*(16), R753–R761. https://doi.org/10.1016/j.cub.2024.06.062

[66]. Smith, O. M., Cohen, A. L., Reganold, J. P., Jones, M. S., Orpet, R. J., Taylor, J. M., Thurman, J. H., Cornell, K. A., Olsson, R. L., Ge, Y., Kennedy, C. M., & Crowder, D. W. (2020). Landscape context affects the sustainability of organic farming systems. *Proceedings of the National Academy of Sciences*, *117*(6), 2870–2878. https://doi.org/10.1073/pnas.1906909117

[67]. Tang, F. H. M., Nguyen, T. H., Conchedda, G., Casse, L., Tubiello, F. N., & Maggi, F. (2024). CROPGRIDS: A global geo-referenced dataset of 173 crops. *Scientific Data*, *11*(1), 413. https://doi.org/10.1038/s41597-024-03247-7

[68]. Valdez, J. W., Callaghan, C. T., Junker, J., Purvis, A., Hill, S. L. L., & Pereira, H. M. (2023). The undetectability of global biodiversity trends using local species richness. *Ecography*, *2023*(3), e06604. https://doi.org/10.1111/ecog.06604

[69]. Van Klink, R., Bowler, D. E., Comay, O., Driessen, M. M., Ernest, S. K. M., Gentile, A., Gilbert, F., Gongalsky, K. B., Owen, J., Pe'er, G., Pe'er, I., Resh, V. H., Rochlin, I., Schuch, S., Swengel, A. B., Swengel, S. R., Valone, T. J., Vermeulen, R., Wepprich, T., … Chase, J. M. (2021). EXPRESSION OF CONCERN: InsectChange: a global database of temporal changes in insect and arachnid assemblages. *Ecology*, *102*(6), e03354. https://doi.org/10.1002/ecy.3354

[70]. Van Klink, R., Bowler, D. E., Gongalsky, K. B., Swengel, A. B., Gentile, A., & Chase, J. M. (2020). Meta-analysis reveals declines in terrestrial but increases in freshwater insect abundances. *Science*, *368*(6489), 417–420. https://doi.org/10.1126/science.aax9931

[71]. Wagner, D. L. (2019). Global insect decline: Comments on Sánchez-Bayo and Wyckhuys (2019). *Biological Conservation*, *233*, 332–333. https://doi.org/10.1016/j.biocon.2019.03.005

8. Appendices

8.1. Appendix 1: Marginal R^2 values reported by other studies using PREDICTS

Table 8-1 Marginal and conditional R^2 reported by references using PREDICTS to answer agricultural land use related questions

Reference	Are conditional and marginal R^2 reported?	Are conditional and marginal R^2 reported discussed in the text?	Marginal R^2	Conditional R^2
Newbold et al 2015	No	No	-	-
Newbold et al 2020	No	No	-	-
Outhwaite et al. 2022a	Yes	No	0.004 to 0.045*	0.49-0.96*
Scheepens et al. 2026	Yes	No	0.000 to 0.084**	0.65 to 0.92**
Fan et al. 2024	Yes	No	0.016 to 0.075	0.89 to 0.98
Ceaușu et al. 2025	Yes	Yes	0.026 to 0.337***	0.49 to 0.98***
Millard et al. 2021	Yes	Yes	0.04 to 0.149	0.70 to 0.88

*Only non-separated (by non-tropical and tropical) models; **Europe's models of total abundance or species richness; ***Crop specific models, only total abundance and species richness

8.2. Appendix 2: R packages used

All data processing, spatial extraction and statistical analysis were carried out in R version 4.5.2 (R Core Team, 2025). Spatial data were handled with *sf* 1.1.1 (Pebesma, 2018) and *terra* 1.8-93 (Hijmans, 2026), and data manipulation and figures with the *tidyverse* (Wickham et al., 2019). Mixed-effects models were fitted with *lme4* 2.0.1 (Bates et al., 2015) and *glmmTMB* 1.1.14 (Brooks et al., 2017); with additional comparison analysis being performed by using penalised variable selection through *glmmLasso* 1.6.4 (Groll, 2026); multi-model inference used *MuMIn* 1.48.19 (Bartoń, 2026); and residual diagnostics used *DHARMa* 0.5.0 (Hartig, 2026).

8.3. Appendix 3: Questionnaire to assess issues in primary literature.

Table 8-2. List of questions used to guide the screening of the original studies. The questions were designed to trigger the identification of potential issues, sometimes beyond the scope of the question itself and enhance by discussions between reviewers.

Topic	*Questions*
Taxa	Is the Reference focusing only on Insecta or Arachnida species or does it also consider other classes?
	Are all groups of Insecta or Arachnida being studied in the Reference reported on the database or is there reason to think than one group is being left out without explanation?
Sites	How many sites are reported by the Reference?
	Is the previous value consistent with the number of sites in the database?
Methodology	Does the reference include sites subject to manipulations or experimentations?
	If so, is it possible to keep some sites (e.g., control sites)? and, is it possible to identify those in the database?
	Is the reference focusing on one very small group of taxa and, therefore should not be accounted for in species richness metrics?
	Does the database contain more than one sampling method?
	If so, Do the groups of SSBS of a same sampling method have a distinct study number?
Counts	Does the Reference provide values of abundance that could be used to check against raw data? are those the full raw values per site or a total across a group of sites?
Coordinates	Does the paper provide coordinates of the sites that could be used to check against raw data if needed?
	Do 2 or more sites share the same coordinates? If so, is there a clear reason for that? (for example, same point measured in more than 1 year)
	Is there is a map of the sites in the Reference that could help check if the sites are approximately in the right place?
Timeseries and temporal assessment	Does the database contain values for multiple years (time series)? are them consistent with the Reference?
	For how long and in what seasons/years did the sampling occur?
	Is it possible that the total values are aggregations of more than one effort/season?
Others	Given all the previous question and any other observation, should all the sites of the reference be kept? If no, which ones should stay? Is it possible to identify them in the database?
	Is there some aspect (coordinates, counts, resolution) that seems to require a modification/correction?

8.4. Appendix 4: Summary of issues found in the primary literature of the dataset

In total, 16 categories of issues across 7 topics were found (Table 1), being the most frequent problem the unavailability of raw data to cross check the information on the database at a site level (37 References), although in many of those cases the total amount of individuals was available and checked. Considering the situations in which total or granular data was available, 22 References showed a difference between values reported in the Source and in the Database for the taxonomic group of interest.

Table 8-3 Number of sources (n) detected with issues in each corresponding issue category. The list of sources is available in supplementary information S1. Numbers in Reference correspond to the numeration in S1.

Topic	Category of issue	n	References (Numeration of Supplementary material S1)
Counts	Difference in site level counts or total abundance values	22	1, 4, 6, 9, 10, 13, 14, 15, 17, 20, 21, 24, 25, 27, 37, 41, 44, 47, 48, 50, 51, 53
Sampling method and effort	Differences in effort potentially not accounted for in corrected data	4	10, 20, 24, 36
	Reference has multiple sampling methods	11	2, 3, 5, 7, 14, 21,30, 32, 34, 37, 38
	Results from different sampling methods are pooled together	3	2, 14, 30
Temporality	Different sites sampled in different years (one year per sample, no time series)	4	10, 20, 24, 30
	Grouped data across seasons and/or years	12	1, 2, 4, 7, 8, 14, 16, 33, 36, 48, 50, 54
	Reference contains time series	7	5, 6, 7, 12, 18, 20, 53
Taxa	Focus on small number of species	5	9, 10, 11, 22, 37
Sites	Inconsistency of number of sites or only partial number of sites reported	6	6, 7, 14, 29, 40, 41
	Shared coordinates for more than one site (reason being different than the existence of multiple sampling methods or time series)	5	5, 28, 39, 44, 47
Insufficient data	Only partial data reported either in the database or in the Reference (different from sites)	11	9, 12, 15, 23, 29, 34, 36, 37, 43, 44, 53
	Raw data is not available in the reference, or it is available but not by site	37	2, 3, 5, 6, 7, 9, 12, 13, 14, 15, 20, 23, 24, 25, 26, 27, 28, 29, 32, 33, 34, 35, 36, 38, 39, 40, 41, 42, 43, 44, 46, 47, 49, 50, 52, 53
Others	References with experimental manipulations	9	6, 12, 27, 31, 32, 35, 46, 49, 53
	Other issues	9	4, 6, 7, 16, 26, 29, 31, 41, 47

The description of all detected issues by reference, and the control action that was taken can be found the Supplementary information S1 (Section 9). In terms of sampling methods and effort, PREDICTS have various variables of interest to describe observations, among them: 'sampling_method', 'sampling_effort', 'rescaled_sampling_effort', and 'Effort_corrected_Measurment'. In four References evidence was found suggesting that a difference in sampling effort might have not been accounted for. In only one of those cases the evidence -rescaled sampling effort being different from sampling effort but effort corrected measurement being equal to the non-corrected measurement- was found in the database. In the rest, rescaled and effort corrected variables do not differ from the base ones despite the literature suggesting that it should be the case. In those cases, it is possible that authors introduced the data having already accounted for the difference in sampling effort, therefore they are only reported. In the first case however, data was corrected by using the rescaled sampling effort values to adjust the effort corrected measurements.

In eight references, more than one sampling method is reported. In one of those occasions, there was a mismatch between the number of methods being reported in the reference (one) and available in the database (two). It was decided for all the cases only to check that different sampling methods within a source have different study numbers. In other four references it was observed that data was pooled from various sampling methods (e.g., pan trapping and sweep netting results). As this is a situation consistent across sites within a study and there was no option to assign individual study numbers, the issues are only reported.

The most common temporal issue was the aggregation of data across seasons or even years (12 studies), which are here only noted, followed by the presence of timeseries in the data of eight sources. Those, will need to be accounted for in the model structure of the analysis that will follow. In the cases were References that contained samples taken in different sites during different years (i.e., some sites being sampled one year, others being sampled in others, different from time series), different study numbers were assigned by year.

Studies focusing on single species were excluded, and the ones focusing on a small number of species or in a single taxon without species level identification were only kept for total abundance analysis but not for species richness analysis (e.g., values reported at family level, only one family).

Regarding sites, inconsistency in sites counts was found in 9 References but it was decided to only reported them without further action as authors might have decided to exclude sites for multiple reasons (e.g., sampling problems, filling effort limitations, lost data, among others). In the cases multiple SSBS having the same coordinates (different from the duplicates due to timeseries or multiple sampling methods), actions were taken case by case as the root cause differ significantly among them. Actions included contacting authors of primary studies to correct coordinates.

Notably, nine sources were found to include different levels and types of manipulations. In four cases, experimentation was done managing small plots in factorials designs with actions including: artificially planting a particular plant species, implementing different mixes of seeds, or application of herbicides. In either of these cases it was possible to identify the control observational sites in the database -or it was only possible to find one which is not enough to maintain a reference as the study that follows will require sites in various habitats within a study- therefore they were excluded. Other 2 sources follow similar issues but it was possible to differentiate a group of sites not subjected to the experimentations and those were kept while excluding the manipulation sites. In the same type of issue, 3 sources were found to include grazing experiments. Given the extent in time and space of the manipulations, and the common nature of these situations, it was decided to keep all sites.

Finally, in nine references other issues not included in the previous categories were found. These are of very different nature. For example, sampling methods not being detailed enough in the database to capture differences reported but the authors or sites with exchanged total counts. The actions taken to manage them were case specific and can be found in the Supplementary information S1.

8.5. Appendix 5: Characterization of the biodiversity dataset and response variable

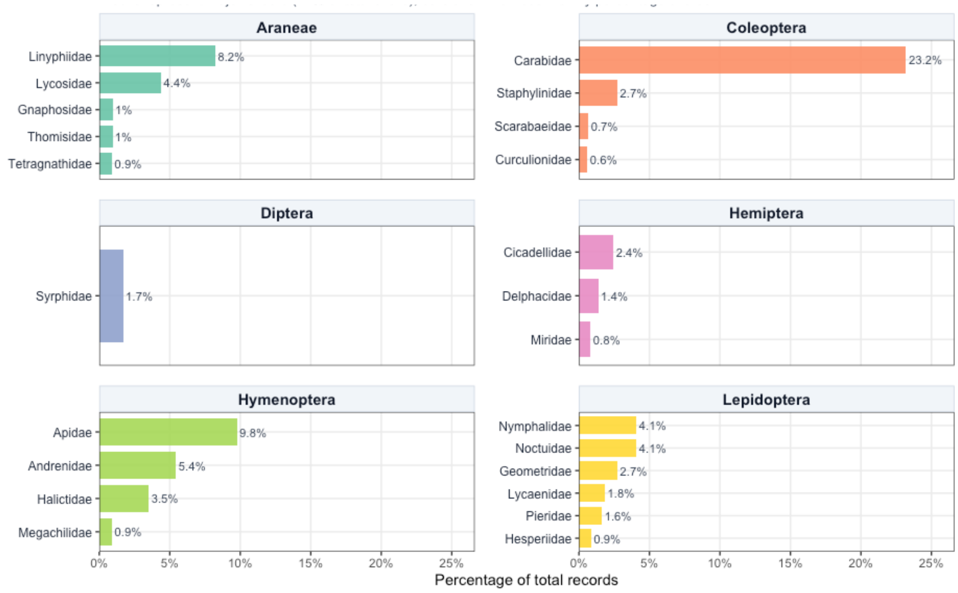


Figure 8-1. Taxonomic composition of the dataset. Bars show individual family percentage shares of total records, with only families with 0.5% of records being shown.

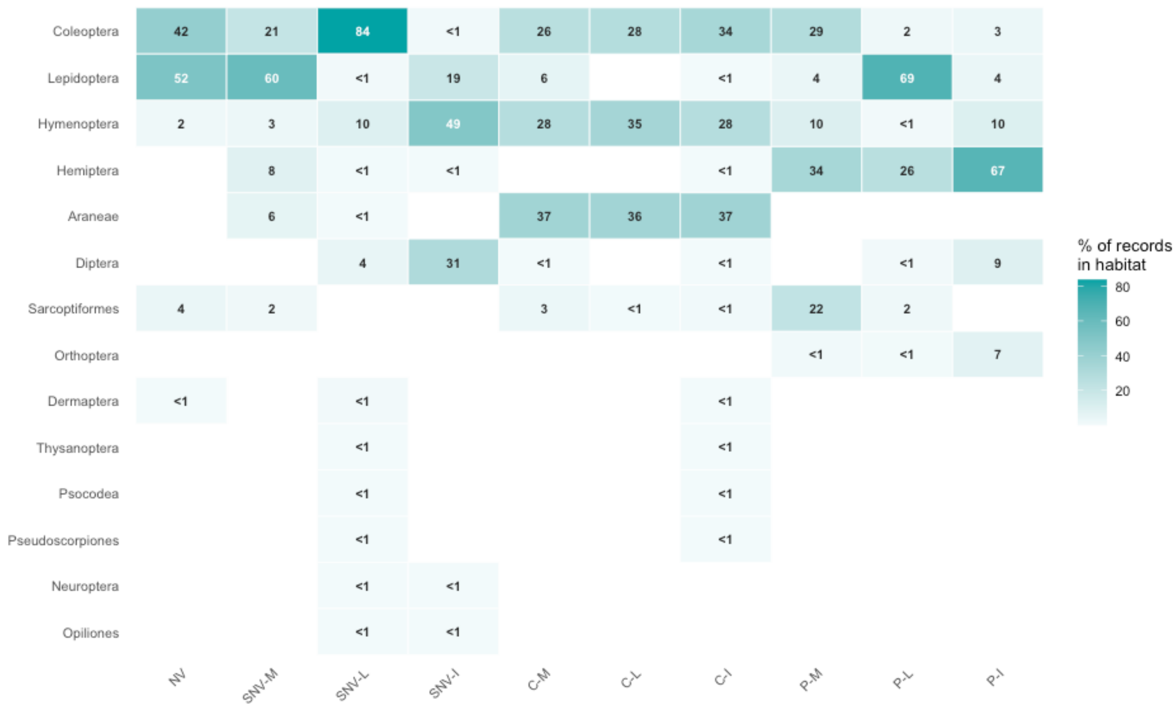


Figure 8-2 Order composition by habitat as percentage of records. Each column sums 100%. Habitat codes are: NV: Natural Vegetation, SNV: Semi-natural vegetation, C: Cropland, P: Pasture, M: Minimal, L: Light, I: Intense.

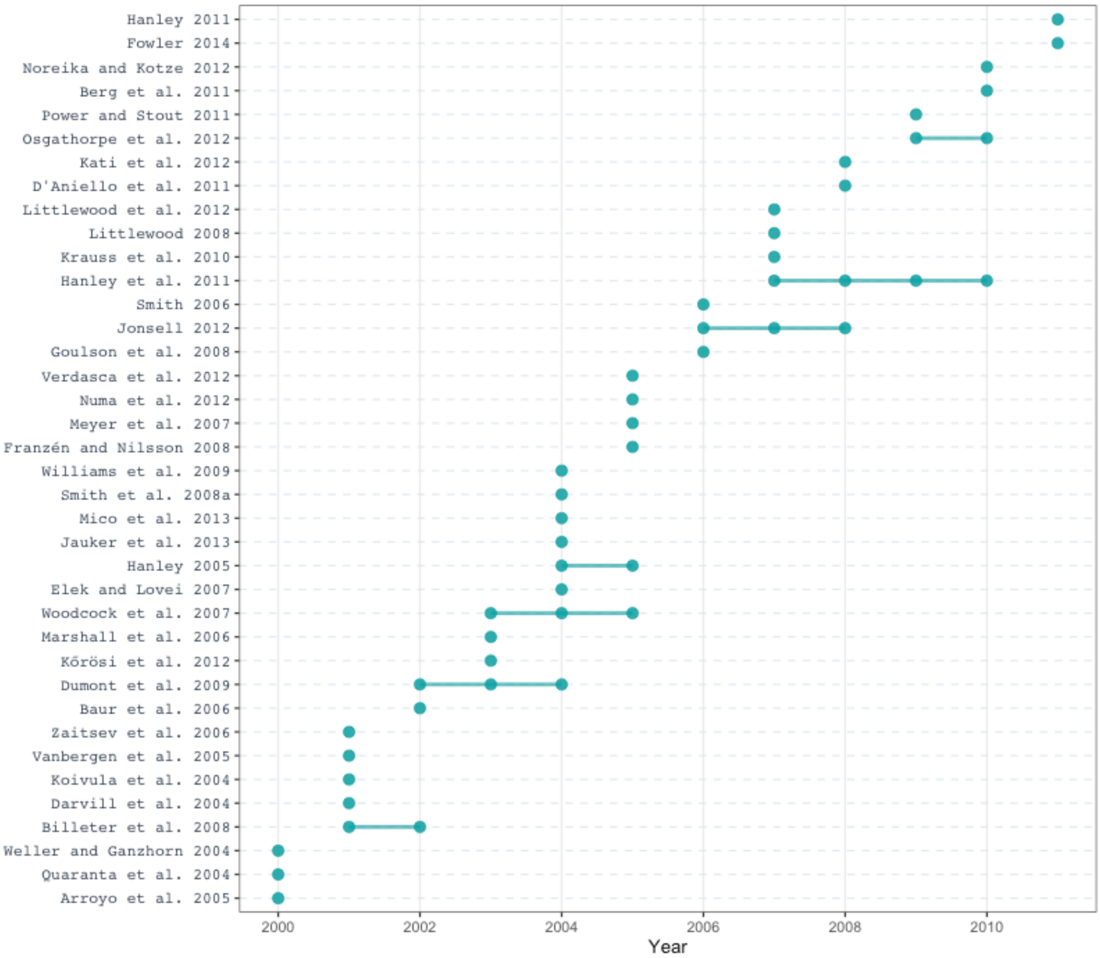


Figure 8-3 Years with sampling events by reference on the dataset. References with consecutive points not correspond necessarily to timelines because some include different sites being sampled in different years.

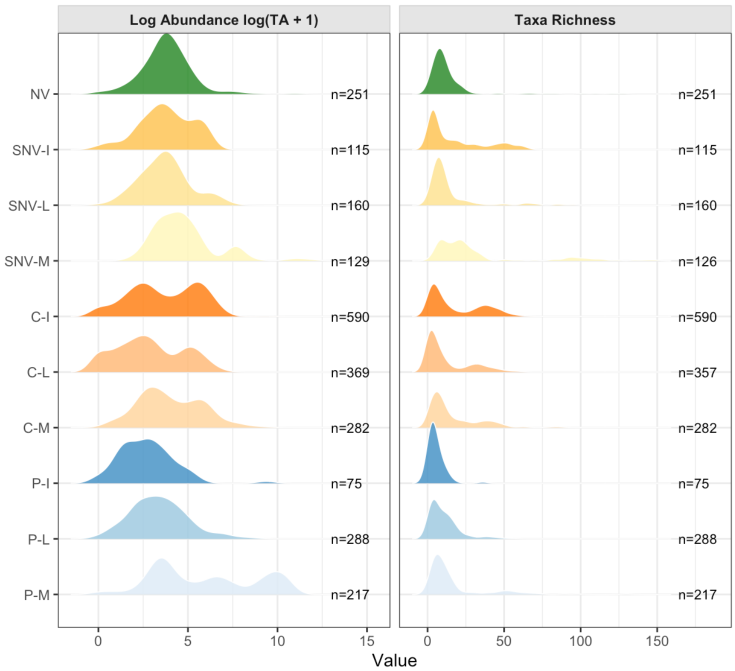


Figure 8-4. Distribution of the response variables by habitat, shown as density plots. For richness, curves are estimated by smoothing over discrete count data. Habitat codes are NV: Natural Vegetation, SNV: Semi-natural vegetation, C: Cropland, P: Pasture, M: Minimal, L: Light, I: Intense. n is the number of sites in each category.

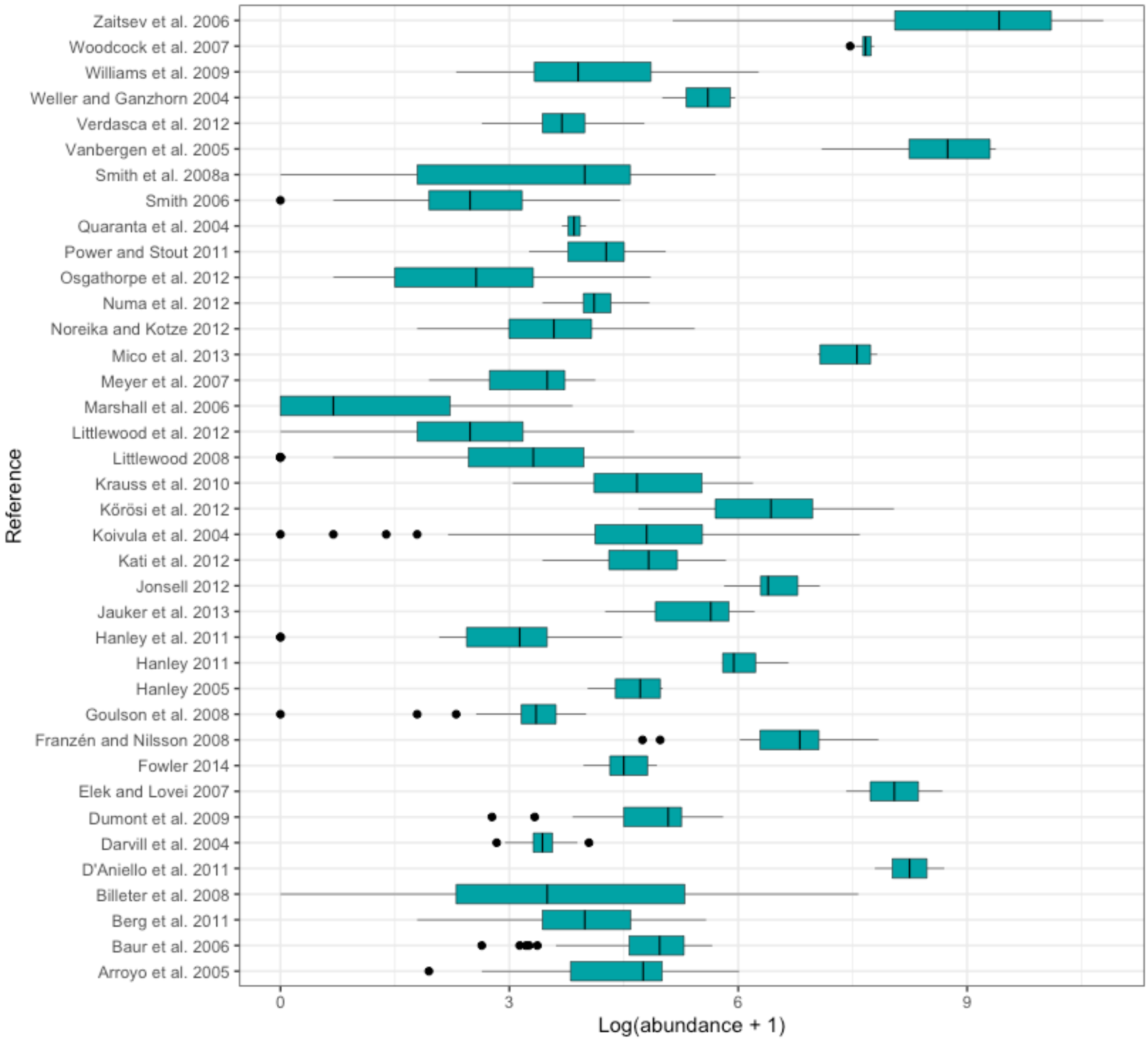


Figure 8-5. Distribution of site level abundance by reference, shown as boxplots. Abundance is log transformed as log(abundance + 1).

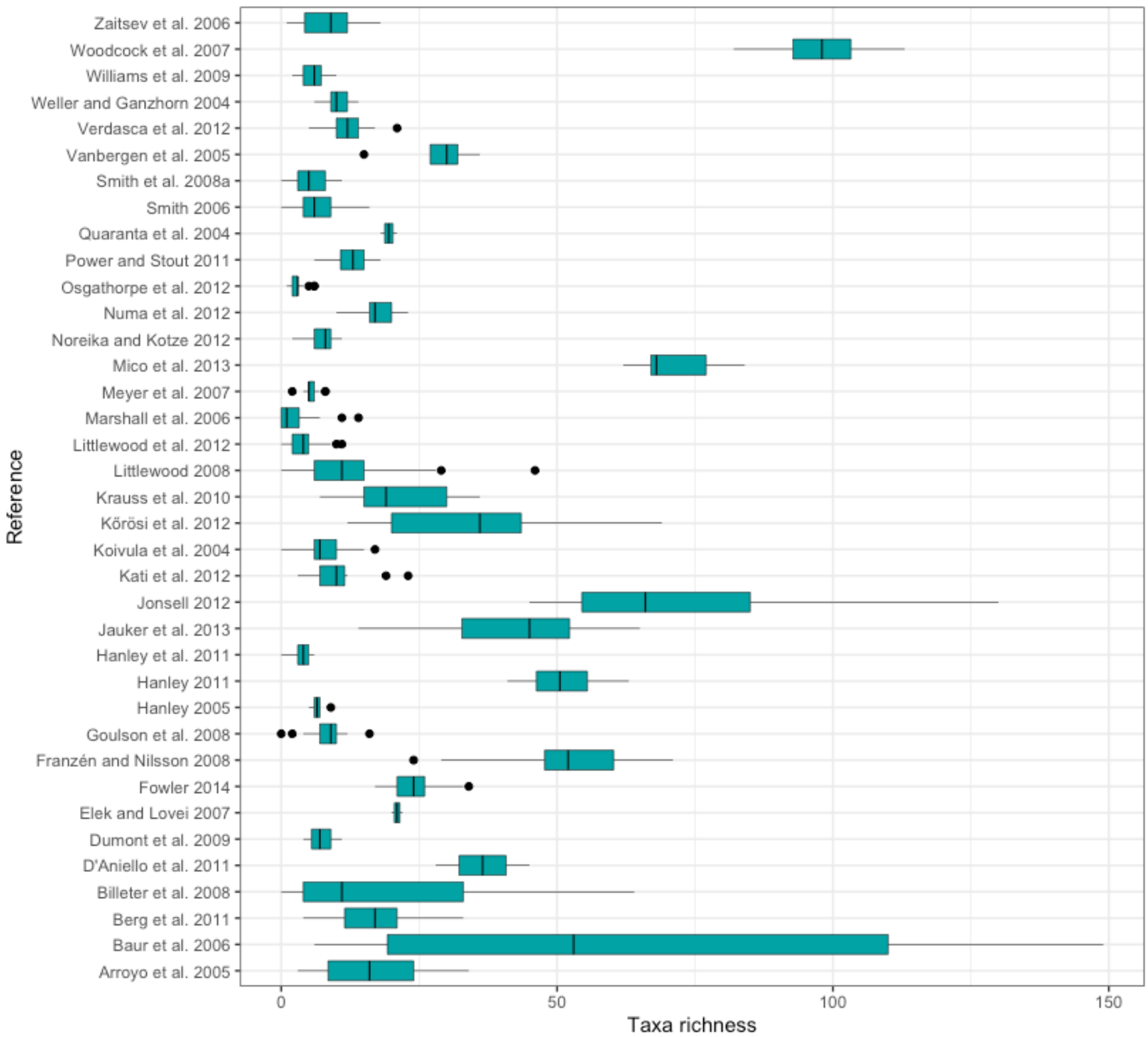


Figure 8-6. Distribution of site level taxa richness by reference, shown as boxplots.

8.6. Appendix 6: List of studies with potential inconsistencies on land use assignment.

Table 8-4 References with potential inconsistencies on land use assignment. Three cases are presented: field margins are assigned a land use different from cropland, grasslands are assigned to multiple land uses and primary vegetation being a sites that might not be undisturbed. The value or values of the "Habitat_as_described" column of the PREDICT database are compared to the subsequent "Predominant_land_use" column.

Reference	Case	Habitat as described by primary authors	Predominant land use given in PREDICTS
Goulson et al. 2008	Field margins are assigned a land use different from cropland	"Pasture or field margin"	"Pasture" or "Young secondary vegetation". The second later transformed by our study in semi-natural vegetation.
Hanley et al. 2011	Field margins are assigned a land use different from cropland	"Perrennial field margin-Wheat", and "Perrennial field margin – Bean"	"Young secondary vegetation" land use, later transformed by our study in semi-natural vegetation.
Smith 2006	Field margins are assigned a land use different from cropland	"Arable field margins"	"Young secondary vegetation" land use, later transformed by our study in semi-natural vegetation.
Smith et al. 2008a	Field margins are assigned a land use different from cropland	"Field margin"	"Young secondary vegetation", later transformed by our study in semi-natural vegetation
Meyer et al. 2007	Grasslands are assigned to multiple different land uses	"Calcareous grassland"	"Secondary vegetation (indeterminate age)", later transformed by our study in semi-natural vegetation.
Kőrösi et al. 2012	Grasslands are assigned to multiple different land uses	"Semi-natural grassland with high grazing pressure"	"Pasture"
Arroyo et al. 2005	Grasslands are assigned to multiple different land uses	"Grasslands/pasture"	"Pasture"
Zaitsev et al. 2006	Grasslands are assigned to multiple different land uses	"Grassland"	"Pasture"
Woodcock et al. 2007	Grasslands are assigned to multiple different land uses	"lowland improved grasslands"	"Secondary vegetation (indeterminate age)", later transformed by our study in semi-natural vegetation.
Littlewood et al. 2012	Grasslands are assigned to multiple different land uses	"Grassland"	"Pasture"
Jauker et al. 2013	Grasslands are assigned to multiple different land uses	"Calcareous grasslands"	"Secondary vegetation" (indeterminate age), later transformed by our study in semi-natural vegetation.
D'Aniello et al. 2011	Grasslands are assigned to multiple different land uses	"Grassland"	"Cropland", or "Pasture"
Kati et al. 2012*	Grasslands are assigned to multiple different land uses	"Mediterranean tall humid herb grasslands of the Molinio-Holoschoenion"	"Primary vegetation", later transformed by our study in natural vegetation, and "Cropland", or "Pasture"
Vanbergen et al. 2005	Primary vegetation sites not being necessarily undisturbed areas	"Forest-dominated mosaic"	"Primary vegetation", later transformed by our study in natural vegetation
Fowler 2014	Primary vegetation sites not being necessarily undisturbed areas	Healthland/Park, Rural	"Primary vegetation", later transformed by our study in natural vegetation
Numa et al. 2012	Primary vegetation sites not being necessarily undisturbed areas	Savanna farmland	"Primary vegetation", later transformed by our study in natural vegetation

*All cases except for this have a value of "Source_for_predominant_land_use" equal to "Direct from publication / author". This case has "Google maps" as described source.

8.7. Appendix 7: Extended methods description of landscape characterization data

The landscape surrounding each sampling site was characterized from publicly available spatial datasets, selected to represent all potential variables affecting the biodiversity response in the targeted habitat categories. Section 2.3 presented a summary of each source and indicator used. Here, an extended description of each indicator if provided.

(i) Percentage of cropland

Cropland cover was obtained from the Global 30-m Annual Cropland Extent Dynamics dataset (GACED30; Liao et al., 2026), which maps cropland extent annually from 2000 to 2024. It was selected for two reasons: its 30 m resolution, the finest available among global cropland products, and its annual coverage, which allows the cropland surface around each site to be matched to the year in which that site was sampled.

GACED30 relies on the seasonal signal of cultivation. The product follows how the reflectance of each 30 meters pixel changes through the year, using Landsat time series, because the sequence of sowing, growth and harvest produces a pattern that distinguishes cultivated land from permanent vegetation. It includes annual herbaceous crops such as cereals and vegetables, perennial woody crops such as orchards, vineyards and plantations, land under active fallow as part of a rotation, and agricultural structures such as greenhouses. It explicitly excludes temporary meadows and pastures, that is, land cultivated with forage for less than five years, due to the difficulty of distinguish it from natural grasslands in satellite imagery.

For each site, the annual layer corresponding to the first year of sampling was selected, and the percentage of the buffer classified as cropland was calculated by area-weighted extraction.

(ii) Percentage of water surface

Inland water and wetland cover were obtained from the High-Resolution Layer Water and Wetness of the Copernicus Land Monitoring Service (European Union's Copernicus Land Monitoring Service information, 2020), a 100 m product for the 2018 reference period, derived from imagery from between 2012 and 2018 water. The layer classifies each pixel by the frequency with which r or wet soil conditions are observed, independently of the vegetation growing on it, and distinguishes five classes: permanent water, temporary water, permanent wet, temporary wet and dry. Sea water, unclassifiable pixels and pixels outside the coverage area are coded separately.

Four water and wetness classes were selected and combined into a single mask for this indicator, the include the following examples (Copernicus Land Monitoring Service, n.d.):

- **Permanent water:** permanent inland lakes, natural and artificial ponds, rivers, permanently flowing channels, and coastal water surfaces such as lagoons and estuaries;
- **Temporary water:** temporary natural and artificial lakes, intermittent rivers, flood areas, water-logged areas, wet agricultural fields including rice fields, and intertidal areas;
- **Permanent wet:** reeds, peat land, and inland and coastal wetlands including salt marshes;
- **Temporary wet:** intermittent wetlands, inland saline marshes, and areas of changing soil moisture.

The variable was calculated as the percentage of the buffer covered by this combined mask. Wetlands are accounted here rather than in the natural vegetation variable of Section 2.3.3.

Sea water is coded as a separate class in the source layer and was excluded from the indicator, which measures inland water and wetness only. However, the percentage of sea water was extracted and saved as a separate indicator for the purposes of assessing the total land use cover of the buffers, process detailed in the following section.

(iii) Percentage of natural vegetation, urban and pasture area

Urban, natural vegetation and pasture cover were obtained from CORINE Land Cover (CLC) (European Union's Copernicus Land Monitoring Service Information, 2020b; European Union's Copernicus Land Monitoring Service Information, 2020c), the 100 m European land cover product of the Copernicus Land Monitoring Service, in its 2020 revision. CLC is produced by visual and semi-automated interpretation of satellite imagery against a standardized 44-class nomenclature.

Two files were used, corresponding to the 2000 and 2006 vintages. Each site was assigned the vintage closest to and not later than its sampling year. Sites sampled between 2000 and 2005 were characterized with CLC2000, and those sampled between 2006 and 2011 with CLC2006. Newer vintages (e.g. 2012 or 2018) are available, however the data from the selected dataset does not extend over 2011.

Three variables were derived, each as the percentage of the buffer occupied by the corresponding classes, calculated by area-weighted extraction:

- **Urban cover** (CLC codes 1–11): continuous and discontinuous urban fabric; industrial or commercial units; road and rail networks; port areas; airports; mineral extraction sites; dump sites; construction sites; green urban areas; and sport and leisure facilities.
- **Pasture cover** (CLC code 18): pastures.
- **Natural vegetation cover** (CLC codes 23–29): broad-leaved forest; coniferous forest; mixed forest; natural grasslands; moors and heathland; sclerophyllous vegetation; and transitional woodland-shrub.

The natural vegetation variable excludes the CLC classes describing open spaces with little or no vegetation such as beaches, dunes and sand, bare rock, sparsely vegetated areas, burnt areas and glaciers because these do not provide the vegetated habitat whose buffering role is being tested.

Total land use was estimated by site by adding the percentage of urban cover, pasture cover, natural vegetation cover, water surface, sea water surface and cropland cover. When the total exceeded 100%, which happens due to the use of different sources for each indicator, the urban, pasture, natural vegetation and water covers were rescaled proportionally, while sea water (data with more reliable and clear limits) and cropland (data with more detailed spatial and temporal resolution) were held fixed.

(iv) Climate conditions

Mean yearly temperature and total annual precipitation were obtained from the ERA5-Land monthly averaged reanalysis (Muñoz-Sabater et al., 2021), produced by the European Centre for Medium-Range Weather Forecasts at a resolution of 0.1°, approximately 11 kilometers at the equator. ERA5-Land is generated by running the European Centre for Medium-Range Weather Forecasts (ECMWF) land surface model with atmospheric forcing downscaled from the ERA5 reanalysis, which yields a continuous record rather than an interpolation of station observations.

Monthly values of 2 meters air temperature and total precipitation were retrieved. For each site, the twelve monthly layers of its sampling year were retrieved. Mean annual temperature was calculated as the mean of the monthly temperature layers and converted to degrees Celsius. Total annual precipitation was obtained by multiplying each monthly value, which ERA5-Land reports as a daily accumulation in meters, by the number of days in the corresponding month, summing across months, and converting to millimeters. Both variables were then extracted as the mean value within each buffer.

(v) Pesticides

Pesticide application rates were obtained from the European pesticide application rate maps of Porta et al. (2025), which cover the member states of the European Union and the United Kingdom at 250 m resolution. The dataset was produced by the authors in three steps. A 250 m crop map for 10 cropping systems (corn, soybean, wheat, cotton, rice, alfalfa, vegetable and fruits, orchard and grapes, pasture and hay, and other crops) was first assembled by combining a high-resolution EU crop map (d'Andrimont et al. 2021) at 10 meters resolution with CORINE Land Cover. The pesticide application rates available as global maps in PEST-CHEMGRIDS (Maggi et al., 2019) were then projected onto that map as a first estimate, and national authorizations and bans in force on 9 January 2018 in Europe were applied. The resulting masses were finally calibrated by country and crop against pesticide use statistics reported by EUROSTAT.

For each active ingredient the dataset provides a low, median and high estimate of the application rate in kg/ha. The median estimate was used here, and a single indicator of pesticide pressure was obtained by summing the median maps of all active ingredients.

This variable should be read as an index of relative pesticide pressure rather than as a complete pesticide load. The authors state that their maps do not include all pesticides authorized in the European Union and therefore cannot be used for an evaluation of total pesticide load. Summing across active ingredients also weights each substance by applied mass alone, without regard to its toxicity to arthropods.

(vi) Nitrogen Fertilization

Nitrogen application was derived from NPKGRIDS (Nguyen et al., 2024), a global dataset of nitrogen, phosphorus and potassium application rates for 173 crops with a reference year of 2020 and a resolution of 0.05°, approximately 5.6 km at the equator. NPKGRIDS was built by combining the crop maps of CROPGRIDS (Tang et al., 2024), a global dataset of cropped area for the same 173 crops, with eight datasets of fertilizer application rates compiled from georeferenced sources and from national and subnational statistics. Data was validated against information provided by FAOSTAT, IFASTAT, and national and subnational statistics organisms, and a data quality map at subnational level was produced based on those comparisons and a quality indicator developed by the authors. As simplification, out of the three nutrients available, only nitrogen was retained and used as proxy for fertilization.

NPKGRIDS reports, for each crop, the nitrogen rate applied per hectare of that crop's harvested area. Because each crop layer has a different denominator, the rates cannot be added across crops. Each crop-specific rate was therefore weighted by the harvested area of the crop taken from CROPGRIDS layers, which shares the grid with NPKGRIDS. The total mass of nitrogen applied in each cell was calculated as the sum across crops of the product of the crop-specific rate and the corresponding harvested area, and divided by the land area of the cell. The resulting variable expresses nitrogen applied per hectare of land, and was extracted as the mean value within each buffer.

(vii) Compositional heterogeneity

Two indicators of the diversity of crops grown in the surrounding landscape were estimated to evaluate compositional heterogeneity.

Shannon crop diversity. The first indicator was computed from CROPGRIDS (Tang et al., 2024), a global dataset reporting, for each of 173 crops, the area harvested in each cell of a 0.05° grid, approximately 5.6 km at the equator, for the reference year 2020. The harvested area layers of all 173 crops were stacked, and for each cell the proportion of total harvested area accounted for by each crop was calculated. The Shannon index was then computed on those proportions using the natural logarithm. The process followed the use of Equation 1 (Ortiz-Burgos, 2016) where H' is the diversity index, p_i is the proportion of each crop in the sample, and ln p_i is the natural logarithm of this proportion. The index equals zero where a single crop accounts for all of the harvested area of a cell, and increases as the harvested area is divided more evenly among a greater number of crops. Cells containing no harvested area were left undefined.

$$H' = -\sum_{n=1}^{n} (pi^{*} \ln\ p_{\mathrm{i}}) \quad \text{Equation 1}$$

Crop alpha diversity. The second indicator was taken from the crop diversity dataset of Machefer et al. (2024), which quantifies crop diversity from a satellite-derived crop map at 10 m resolution for 2018. At 1 km × 1 km cells, it provides the 'effective number of crops', defined by the authors as local, or alpha crop diversity. It is estimated by computing the Shannon index at each 1 km cell based on the information of 10-m resolution and then exponentiating that value. The measure is not the count of crop types but the number of equally abundant crop types that would produce the observed diversity, so a cell dominated by a single crop with small amounts of others has a value close to one, while a cell with several crops in similar proportions scores close to the number of types present. The mean α-diversity within each buffer was extracted. A value of zero indicates that no cropland is present in the cell.

The main difference between both indicators is the resolution, the sources, the baseline years and the final exponentiation of the index, which the authors explain "creates equivalence classes among communities, converting entropy into effective number of species" (Machefer et al., 2024, p.2). A collinearity assessment was later performed between the predictors to select the most relevant non-correlated fixed effects for the model (Section 2.4).

(viii) Configurational heterogeneity

Field size was obtained from the global field size dataset of Lesiv et al. (2019), produced through a crowdsourcing campaign run in 2017 in which participants visually interpreted high resolution satellite imagery from Google Maps and Bing through the Geo-Wiki application. Fields were defined as enclosed agricultural areas, including annual and perennial crops as well as pastures, hayfields and fallow land, and were assigned to one of five size classes, from very small to very large. The resulting product has a resolution of 0.005°, approximately 550 m at the equator.

The interpretation of this layer requires care. Each cell records only which field-size class is dominant within it, not the distribution of field sizes it contains and not the size of any individual field. There are 5 field sizes: very small fields (area < 0.64 ha), small fields (between 0.64 and 2.56 ha), medium fields (between 2.56 and 16 ha), large fields (between 16 and 100 ha) and very large fields (>100 ha). A cell classified as small fields is one in which small fields are the most frequent class, and it may still contain fields of other sizes.

Two variables were derived, each as the percentage of the buffer occupied by cells of the relevant classes, calculated by area-weighted extraction:

- **Small fields prevalence:** percentage of the buffer where small (between 0.64 and 2.56 ha) or very small fields (<0.64 ha) are dominant;
- **Large field prevalence:** percentage where large (between 16 and 100ha) or very large fields (>100ha) are dominant.

These are measures of the extent of landscape dominated by a given field-size class, rather than of mean field size. Cells recording the absence of fields were excluded from the denominators, and medium field pixels were not used for the indicators.

Other indicators of configurational heterogeneity were also computed using the landscapemetrics R package (Hesselbarth et al., 2019), identified by Dutt et al. (2026) as one of the tools that allow fragmentation metrics to be produced within reproducible analytical pipelines. These included clumpiness indices for cropland and vegetation which describe how disaggregated or aggregated each particular land use class is. They were not retained, because the indices are undefined for a class absent from a buffer, which produced missing values at a large number of sites and would have reduced considerably the sample available for modelling. They are therefore not described further.

8.8. Appendix 8: Null model structures

Table 8-5 Null models structures and AIC results

Response variable	Model Structure	Package	AIC
log (Abundance + 1)	~ 1 + (1|SS) + (1 | SSB)	lmer()	6518.0
log (Abundance + 1)	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID)	lmer()	6491.9
log (Abundance + 1)	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID) + (1 | Sample_start_year)	lmer()	6491.4
log (Abundance + 1)	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID) + (1 | Sampling_method)	lmer()	6484.9
log (Abundance + 1)	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID) + (1 | Sample_start_year) + (1 | Sampling_method)	lmer()	6485.1
Taxa richness	~ 1 + (1|SS) + (1 | SSB)	glmmTMB()	14141.2
Taxa richness	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID)	glmmTMB()	14132.0
Taxa richness	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID) + (1 | Sample_start_year)	glmmTMB()	14131.4
Taxa richness	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID) + (1 | Sampling_method)	glmmTMB()	14129.9
Taxa richness	~ 1 + (1 | SS) + (1 | SSB) + (1 | Coordinate_ID) + (1 | Sample_start_year) + (1 | Sampling_method)	glmmTMB()	14129.4

8.9. Appendix 9: Characterization of predictors

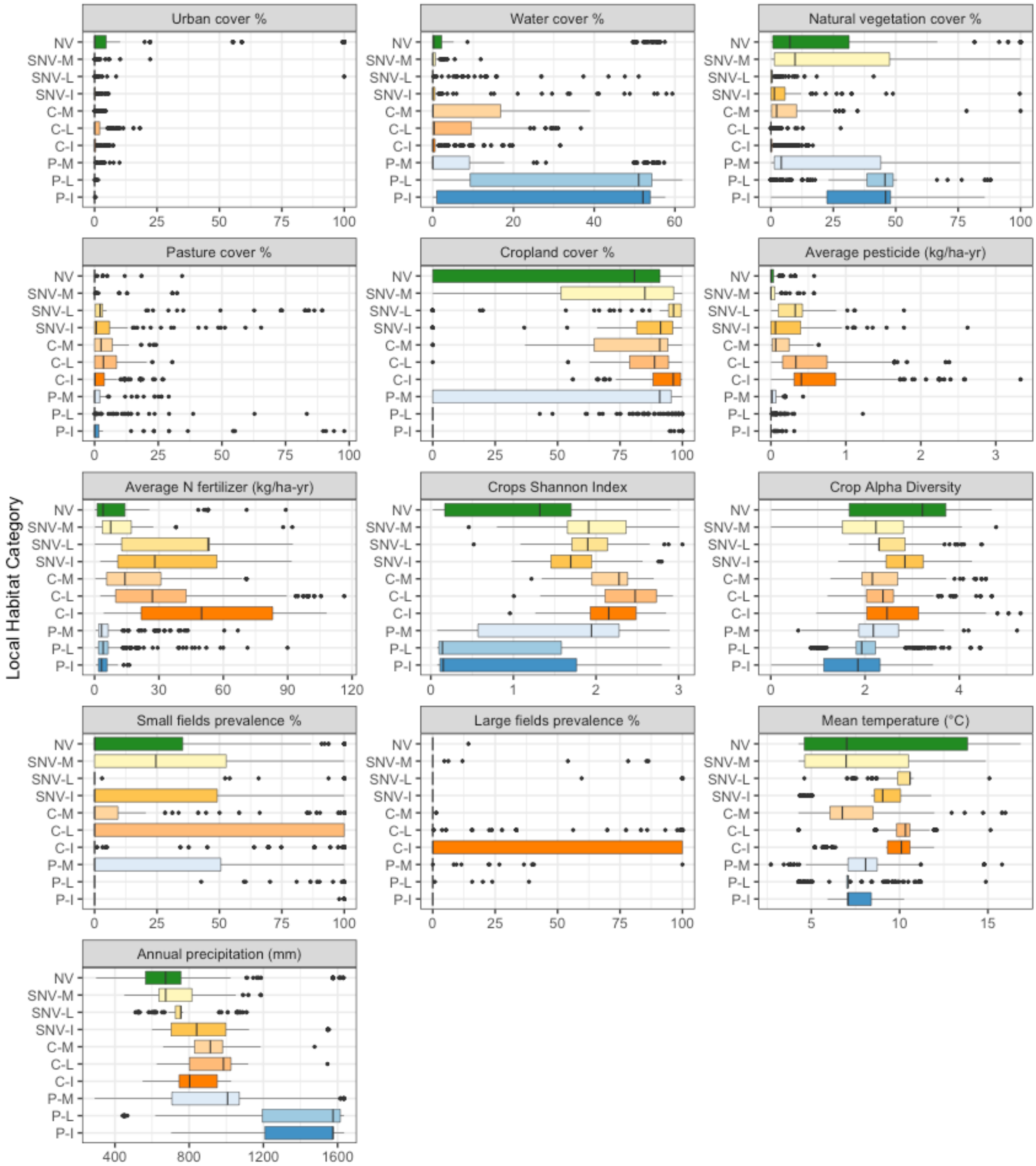


Figure 8-7. Distribution of landscape indicators by habitat category, shown as boxplots. Habitat codes are: NV: Natural Vegetation, SNV: Semi natural vegetation, C: Cropland, P:Pasture, M:Minimal, L:Light, I:Intense.

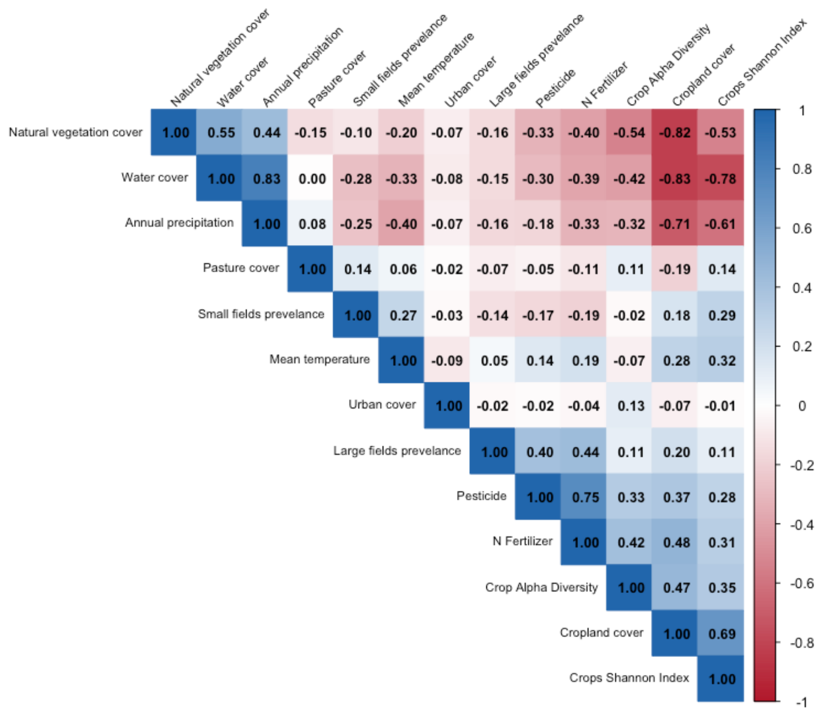


Figure 8-8 Correlation matrix of predictors. Negative relations are presented in a red gradient while positive relations in a blue gradient. Plot obtained using the corrplot() R package.

Table 8-6 Mean and standard deviation of each predictor in its original units, computed over the sites included in each model. Model coefficients (Section 3) are expressed as the change in the response per one standard deviation increase in the predictor. Only predictors included in the global model are listed.

Predictor	Name in figures	Abundance model		Richness model		Unit
		Mean	Standard deviation	Mean	Standard deviation	
Percentage of water surface	Water cover	10.31	18.74	10.37	18.79	% of buffer area
Percentage of urban surface	Urban cover	2.21	11.86	2.21	11.89	% of buffer area
Percentage of natural vegetation surface	Natural vegetation cover	12.23	21.78	12.27	21.83	% of buffer area
Percentage of pasture surface	Pasture cover	3.74	9.41	3.74	9.43	% of buffer area
Average total pesticide application rate	Pesticide use	0.31	0.45	0.31	0.46	kg/ha-year
Average Alpha Diversity of Crops	Crop diversity	2.43	0.82	2.43	0.82	Effective number of crops
Percentage of surface where small and very small fields dominate	Small field prevalence	21.48	38.31	21.62	38.39	% of buffer area
Percentage of surface where large and very large fields dominate	Large field prevalence	8.07	26.44	7.63	25.79	% of buffer area
Mean temperature	Mean temperature	8.79	2.57	8.78	2.58	°C

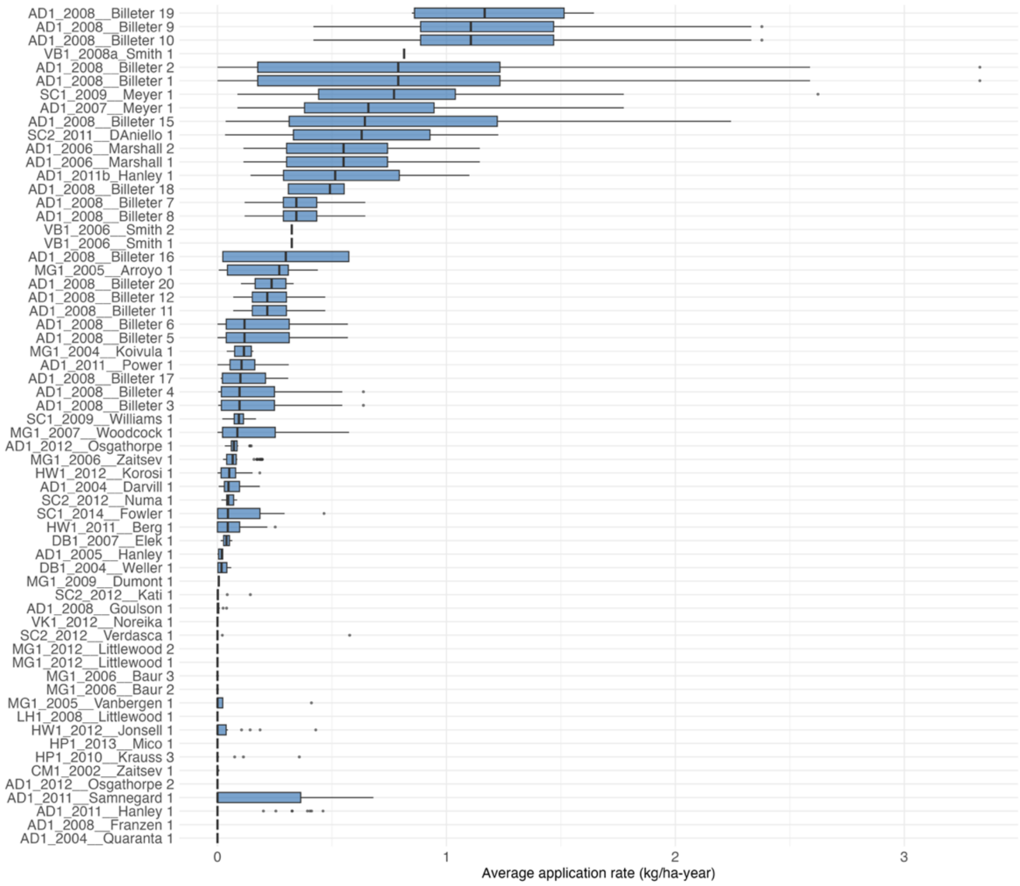


Figure 8-9. Distribution of Average Pesticide Application Rate (kg/ha-yr) per study. Pesticides correspond to one of the significant predictors in the models.

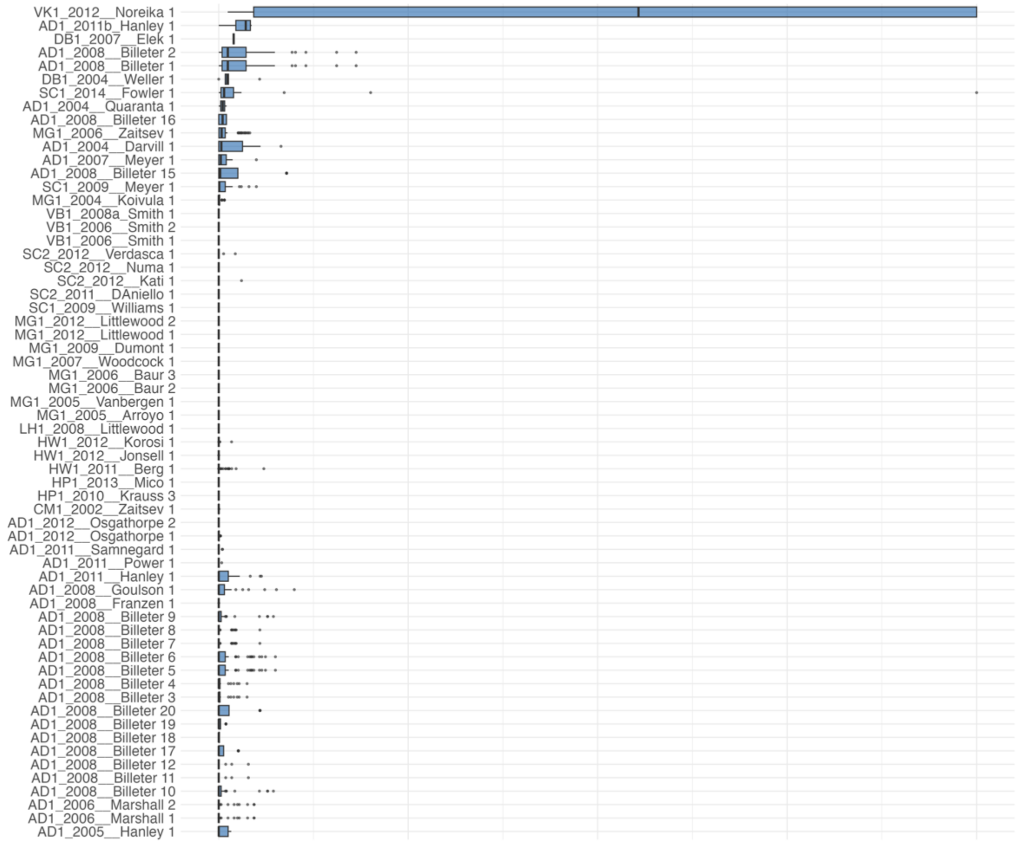


Figure 8-10. Distribution of Urban cover per study. Urban cover resulted a significant predictor in the abundance models

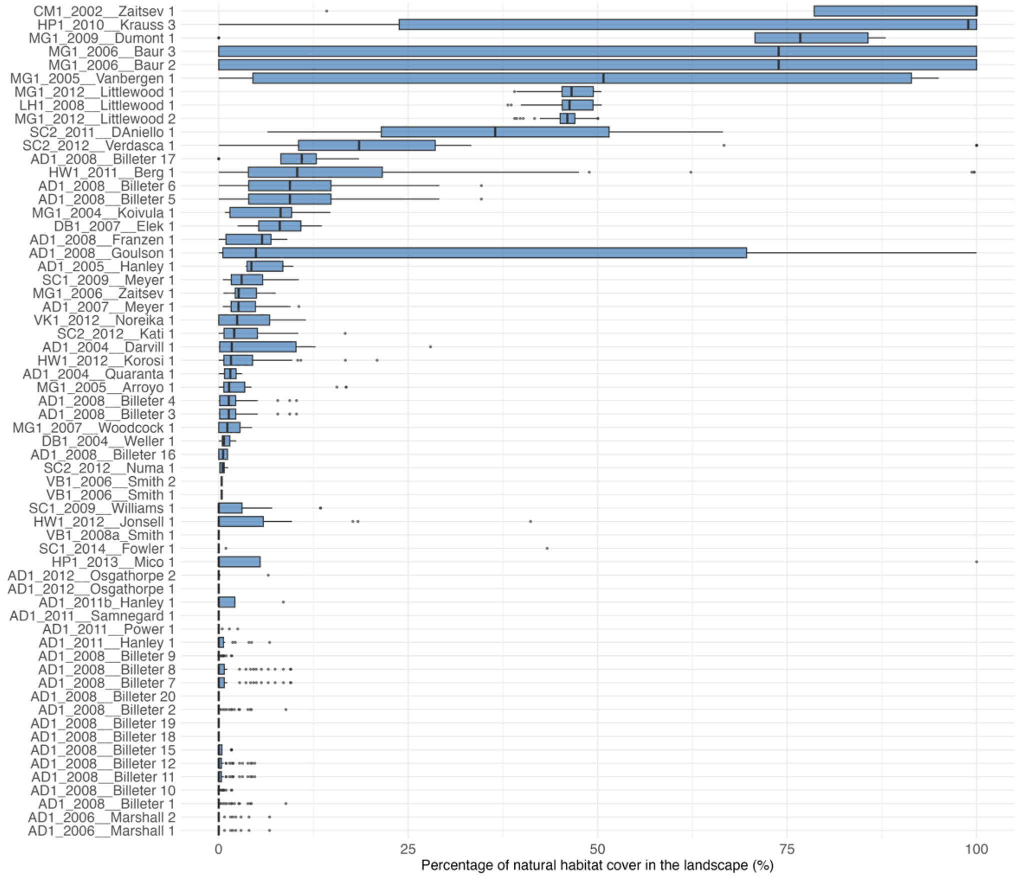


Figure 8-11. Distribution of natural cover per study. Natural cover resulted a significant predictor in the richness models

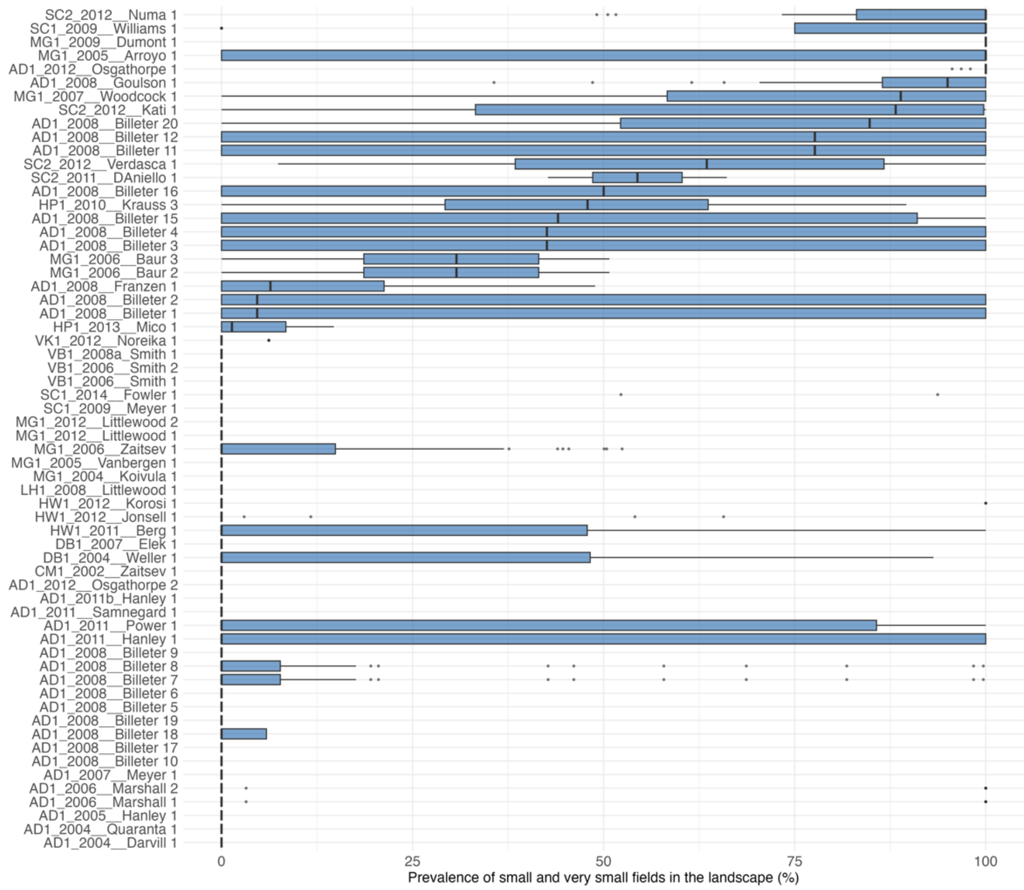


Figure 8-12. Distribution of small field prevalence per study. It resulted a significant predictor in both models

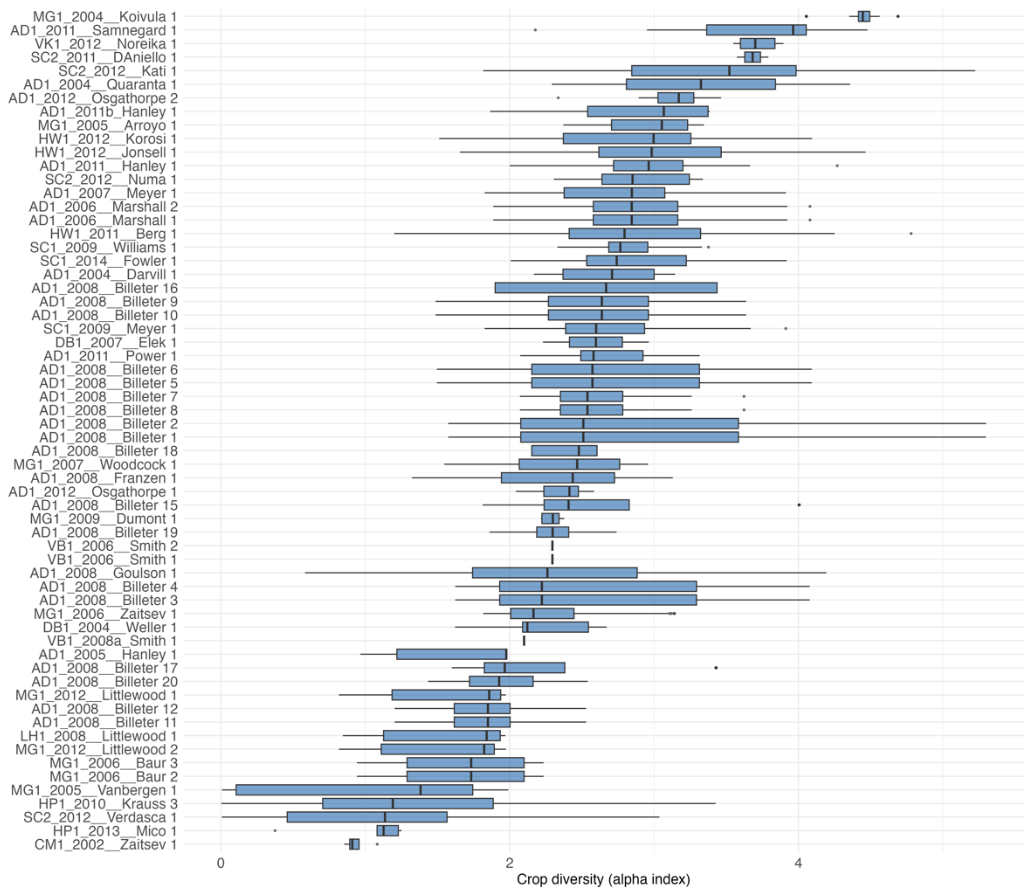


Figure 8-13. Distribution of crop diversity per study. It resulted a significant predictor in both models

8.10. Appendix 10: DHARMa plots of the best performing abundance and taxa richness models

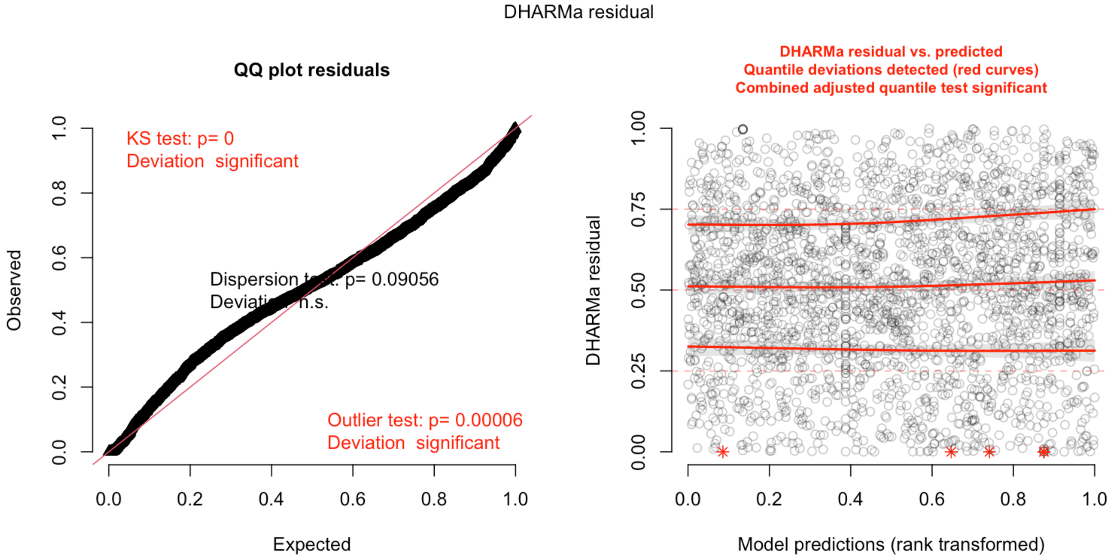


Figure 8-14 DHARMa residual diagnostics, abundance model (Gaussian, best model). Residual diagnostics for the best-supported model of total abundance (log-transformed), based on n = 12500 simulations, generated with the DHARMa package. Left: quantile-quantile plot of scaled residuals against the expected uniform distribution, with the Kolmogorov-Smirnov test for overall deviation from uniformity and tests for outliers and dispersion. Right: scaled residuals plotted against rank-transformed model predictions, with quantile regression lines (red, at the 0.25, 0.5 and 0.75 quantiles) compared against their expected horizontal positions (dashed) to detect any systematic pattern in residual spread across the range of fitted values. Best-supported model correspond to the model with best AIC after running the dredge function.

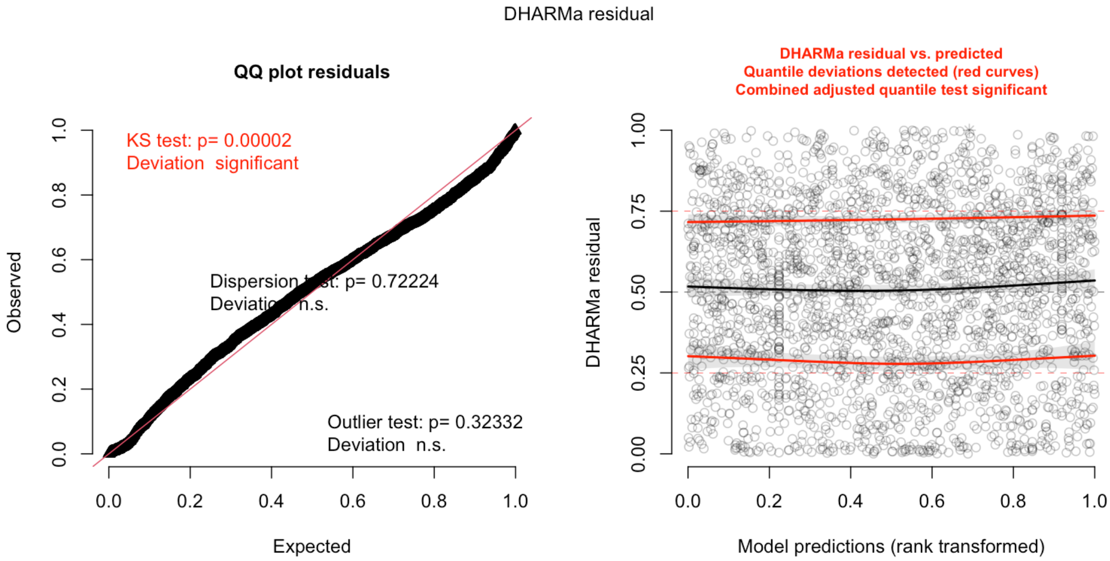


Figure 8-15 DHARMa residual diagnostics, richness model (zero-inflated negative binomial, best model). Scaled residual diagnostics for the best-supported model of taxa richness (zero-inflated negative binomial), based on n = 12500 simulations, generated with the DHARMa package. Left: quantile-quantile plot of scaled residuals against the expected uniform distribution, with the Kolmogorov-Smirnov test for overall deviation from uniformity and tests for outliers and dispersion. Right: scaled residuals plotted against rank-transformed model predictions, with quantile regression lines (red, at the 0.25, 0.5 and 0.75 quantiles) compared against their expected horizontal positions (dashed) to detect any systematic pattern in residual spread across the range of fitted values. Best-supported model correspond to the model with best AIC after running the dredge function.

8.11. Appendix 11: Penalized regression as complementary analysis

Because the number of predictors and interactions considered is large, a second approach to arriving at a final model was tested. The method used was the LASSO approach, a penalized regression in which a single model containing all the predictors is fitted and a penalty proportional to the size of the coefficients is added to the likelihood, instead of fitting every combination of predictors and ranking them. The penalty pulls all coefficients towards zero, so that selection and estimation happen in one step. Groll and Tutz (2014) extended the LASSO to generalized linear mixed models, allowing the random effects structure to be kept, and provided a glmmLASSO R package (Groll, 2026) which is use here for the modelling.

In this approach the predictors do not need to be filtered for correlation beforehand, since where two variables carry similar information, the penalty retains one and shrinks the other towards zero. The full set of landscape predictors was therefore used, including cropland cover, nitrogen fertilization, Shannon crop diversity and annual precipitation, which had been removed from the main models.

Two model structures were implemented. For abundance, a log transformed response variable was used (log(ab+1)) with a gaussian family distribution. For richness, the situation is more complex as the glmmLASSO () package does not support a zero inflated negative binomial distribution, nor a negative binomial distribution. The package only supports the family arguments that come from the glm () function of the stats R package, which include: binomial, gaussian, gamma, inverse.gaussian, poisson, quasi, quasibinomial an quasipoisson. Given this limitation, the more straightforward options are: (a) using the mpath R package that includes negative binomial families but that doesn't support a structure with random effects, or (b) keeping the random effect structure with using a Poisson distribution. Given the importance the random structure has in the data, as proved by the portion of the variance explain by the random factors in the mixed effects models that were used previously, the second approach is here use and an observation level random effect is added to account for some of the overdispersion.

In LASSO penalized regressions, the strength of the penalty is controlled by a parameter called lambda (λ). The larger its value, the stronger the shrinkage and the fewer predictors are retained. Lambda was selected by fitting each model (abundance and richness) over a range of values from 0[2] to 2000 with steps of 50, and computing the Bayesian information criterion (BIC) for each, following the approach used by Schelldorfer et al. (2011).

As observed in Figure 8-16, the BIC curve for abundance falls steeply and then remain almost flat, with its minimum at a high value of lambda (λ=1650). Such lambda results in very few predictors surviving, therefore a second value of lambda is also tested, corresponding to the first marked drop in the curve (λ=500).

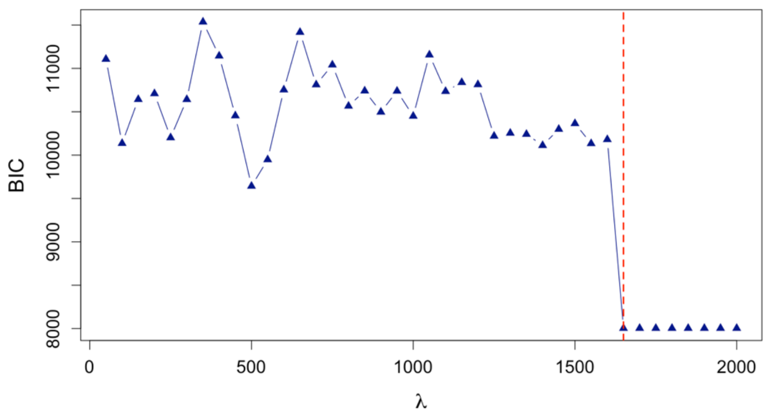


Figure 8-16 Bayesian information criterion change by lambda in the abundance LASSO penalized regression model. Lambda was estimated with 50 units steps. Red dotted line shows the minimum lambda.

[2] Fort he abundance model, it started from 50, since a lambda of 0 could not converge.

In the case of richness, given the optimal value was already in values lower to 500, and additional step of iteration between lambda 300 and 500 was performed with steps of 10 units, results shown in the figure below. Minimum optimal lambda was 380.

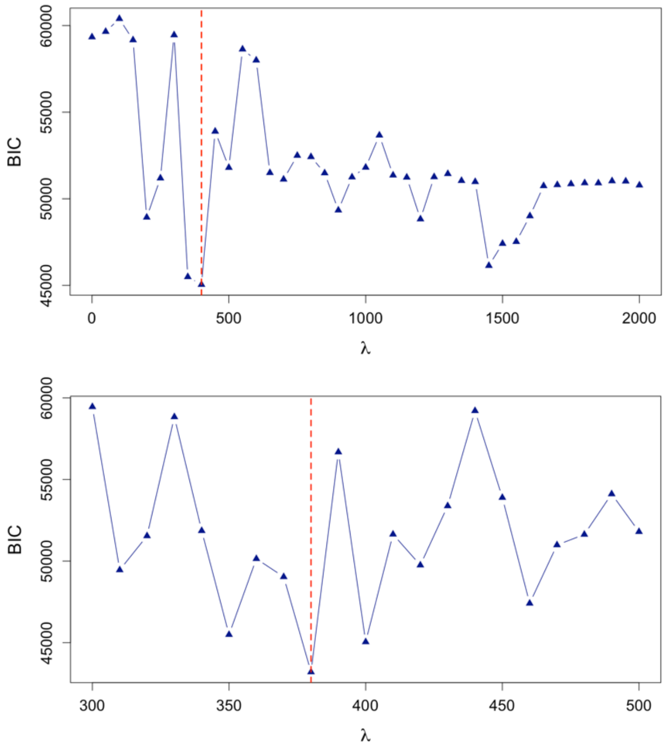


Figure 8-17 Bayesian information criterion change by lambda in the taxa richness LASSO penalized regression model. Up: Lambda estimated with 50 units steps until lambda 2000, Down: Lambda estimated with 10 units steps between lambda 300 and 500. Red dotted line shows the minimum lambda in each.

After choosing lambda values, models were run (abundance lambda 1650, abundance lambda 500 and richness lambda 380), and results were extracted. Three limitations should be kept in mind. First, the package does not return standard errors for the penalized coefficients, so the output shows which predictors are retained without attaching a measure of uncertainty. Second, the coefficients are shrunk towards zero by construction, so their magnitudes are not comparable with those of the unpenalized models studied in the main models of this work. Third, the package does not support the r2() function from the performance package that has been used previously for obtaining marginal and conditional R^2, so values are estimated manually using the Nakagawa & Schielzeth (2013) formulas (equations 26 and 30 of the original document) -same as the one used in the performance package- based on the fixed, random and residual variances of the models.

Model fit was inspected by looking at the DHARMa plots of the three models. Since the simulateResiduals() options is only defined for lmer an glmmTMB models (no glmmLASSO models), residuals were simulated manually with 12500 simulations (based on recommendation explained as footnote in section 2.4.4), then the createDHARMa() function was used to plot.

As is shown in Figure 8-18, both abundance models and the richness model show statistically significant deviations from uniformity (KS test), though the abundance models remain reasonably well-behaved overall when compare to the richness models (lambda=1650 has a flatter residual trend than lambda=500 but more outlier excess). The richness model shows severe non-uniform deviation and dense outlier clustering at both extremes (0 and 1), consistent with the overdispersion of the data which was not then captured by the observational level entity added in the random effects. As explained before, in the main non penalized models a zero inflated negative binomial distribution was used but it was not available for lasso, therefore a poison distribution with observation level entity was tried).

Given these findings, only the results of the abundance optimal model (lambda=1650) is presented below; full results for all three glmmLasso models are nonetheless Supplementary Information S2.

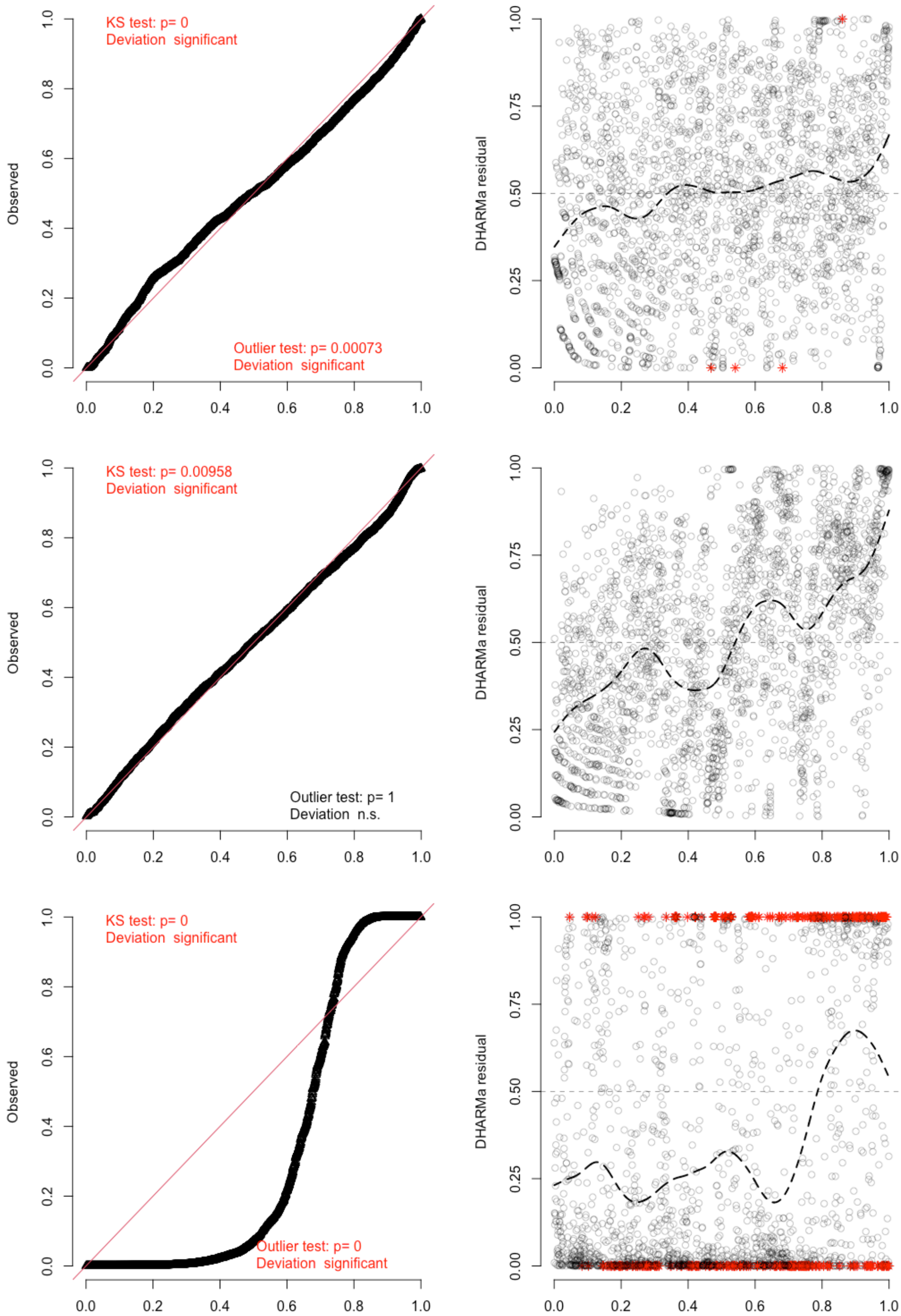


Figure 8-18 DHARMa residual diagnostics for glmmLASSO abundance and richness models. Top: total abundance Gaussian model residuals (log-transformed) with optimal lambda = 1650. Middle: total abundance Gaussian model residuals (log-transformed) with lambda = 500. Bottom: taxa richness Poisson model residuals with optimal lambda = 380. Left: quantile-quantile plot of scaled residuals against the expected uniform distribution, with the Kolmogorov-Smirnov test for overall deviation from uniformity and a test for outliers. Right: scaled residuals plotted against rank-transformed model predictions, with a smooth spline fit to the residuals compared against their expected horizontal position at 0.5 (dashed line) to detect systematic patterns in residual spread. In the latest, quantreg option was set off.

Similar to the non-penalized models, the optimal lambda abundance showed a high conditional R^2 of 0.74 and a very low marginal R^2 of 0.0071. Table 8-7 shows the results of the response to land use and use intensity. As expected with such a high lambda value, the number of predictors retained in the model was small. Only the cropland effect is retained with minimal and intense use having a small negative impact in abundance compare to natural vegetation (-4.07% and -1.50%), and lightly use cropland hosting a small positive effect (+3.92%). As it was seen in the non-penalized models, a clear intensity gradient is not observed.

Table 8-7 Responses of total abundance to local habitat category and use intensity, relative to natural vegetation. Results are presented as percentage change (%), obtained using the glmmLASSO R package with a BIC-optimal lambda of 1650.

Local Habitat	Use intensity	Total Abundance Model (%) λ=1650
Semi natural vegetation	Minimal	0
	Light	0
	Intense	0
Pasture	Minimal	0
	Light	0
	Intense	0
Cropland	Minimal	-4,07
	Light	3,92
	Intense	-1,50

Similarly, most of the landscape related indicators were shrunk to zero with a lambda of 1650. (Table 8-8). Only climatic conditions (temperature and precipitation) showed an effect in abundance, with increase in temperature hosting 11.65% more individuals and precipitation 0.12%. Additionally, two variables showed one relevant interaction. According to this model, pesticides reduce in 0.54% the abundance of insects and arachnids in intensively used croplands and increase of small field prevalence in the landscape increases total abundance by 39% in lightly used pastures.

Table 8-8 Responses of total abundance to landscape composition and agricultural practices with Lasso regression model. Results are presented as percentage of change (%), obtained using the glmmLASSO R package with a BIC-optimal lambda of 1650. Interactions are presented when they were retained.

Landscape composition and Agricultural practices	Total Abundance Model (%) λ=1650	Interactions
Urban cover	0	-
Water cover	0	-
Natural vegetation cover	0	-
Pasture cover	0	-
Cropland cover	0	-
Pesticides	0	-0.54% for intensively used croplands
N Fertilization	0	-
Crop diversity Shannon Index	0	-
Alpha Crop Diversity	0	-
Small fields prevalence	0	+39% for lightly used pasture
Large fields prevalence	0	-
Mean Temperature	11,65	-
Total precipitation	0,12	-

Compared with the model-averaged results reported in Section 3, the LASSO abundance model retains a narrower set of effects, and where a predictor is retained, its direction and magnitude often diverge from the model-averaged estimate. In the case of the habitats effects model averaging found pasture and semi-natural vegetation to differ substantially from natural vegetation, while the LASSO penalty shrank both habitats to zero, retaining only the cropland contrast, and even reversed its sign at light use intensity (+3.92% under LASSO against −34.9% under model averaging). A similar difference is seen for landscape composition and agricultural practices. Pesticide application, which reduced abundance by 12.9% (p=0.002) with no evidence of habitat dependence in the main analysis, was retained by LASSO only as a small interaction specific to intensively used cropland (−0.54%), and urban cover, which reduced abundance in the main model, was dropped in LASSO. The interaction retained for small field prevalence differs as well. The main analysis found significant increases in natural vegetation (29.3%) and intensely used pasture (29.0%) alongside a decrease in minimally used pasture (−19.0%), whereas LASSO instead retained only lightly used pasture (+39%) that was not significant in the main analysis. Conversely, LASSO retained mean temperature (+11.65%) and precipitation (+0.12%) as effects. The first was not significant in the model-averaged and the second could not be tested due to high correlation with other variables. Given that the penalized coefficients do not provide standard errors and that the optimal lambda estimated based on BIC left only a small number of terms, these differences are a clear consequence of LASSO retaining coefficients that are largest in a single model fit, rather than checking which structure provides the best explanation of the data.

8.12. Appendix 12: Estimates, confidence intervals and p values of results

Table 8-9 Estimates, Confidence Intervals and p-values of Abundance results

Indicator	Mean Estimate (%)	Low CI Value (%)	High CI Value (%)	p-value
Values Figure 3-1				
Natural vegetation	0	0	0	-
Cropland Intense	-22,85	-43,70	5,71	0,1066507
Cropland Light	-34,92	-53,97	-7,98	0,0151224
Cropland Minimal	-23,73	-43,61	3,17	0,0789989
Pasture Intense	-28,51	-50,81	3,89	0,0785485
Pasture Light	23,20	-1,64	54,32	0,0695563
Pasture Minimal	96,24	51,01	155,03	0,0000005
Semi-natural vegetation Intense	-3,71	-34,81	42,23	0,8492831
Semi-natural vegetation Light	72,41	21,32	145,01	0,0023954
Semi-natural vegetation Minimal	18,01	-13,01	60,09	0,287547
Values Figure 3-2				
Crop diversity	-16,04	-27,17	-3,21	0,0160257
Natural habitat cover	-5,55	-12,97	2,51	0,1716873
Small field prevalence	29,27	7,40	55,60	0,006658
Pesticide use	-12,93	-20,22	-4,98	0,001911
Urban cover	-9,02	-16,37	-1,01	0,0281135
Large field prevalence	4,07	-6,85	16,28	0,4807825
Pasture cover	0,45	-2,42	3,41	0,7624664
Mean temperature	2,07	-11,25	17,38	0,7742288
Water cover	0,63	-6,04	7,77	0,8577035
Values Figure 3-3 – Crop diversity				
Natural vegetation	-16,04	-27,17	-3,21	0,015974238
Cropland Intense	24,38	7,93	43,35	0,002579826
Cropland Light	17,96	-0,56	39,93	0,058092558
Cropland Minimal	17,26	-1,13	39,07	0,067251108
Pasture Intense	-22,14	-42,44	5,33	0,104545399
Pasture Light	2,93	-12,95	21,71	0,735320342
Pasture Minimal	-12,56	-25,72	2,93	0,106670265
Semi-natural vegetation Intense	-2,94	-24,46	24,71	0,815567783
Semi-natural vegetation Light	-17,76	-37,51	8,23	0,162868402
Semi-natural vegetation Minimal	-15,87	-28,81	-0,59	0,042459365
Values Figure 3-3 – Small field prevalence				
Natural vegetation	29,27	7,40	55,60	0,006630764
Cropland Intense	2,91	-9,78	17,39	0,669187252
Cropland Light	3,25	-8,72	16,79	0,610833927
Cropland Minimal	3,08	-15,73	26,10	0,767738188
Pasture Intense	28,98	1,44	63,99	0,037838486
Pasture Light	16,90	-1,63	38,91	0,076093294
Pasture Minimal	-18,99	-32,27	-3,09	0,02125155
Semi-natural vegetation Intense	3,69	-13,80	24,73	0,700840845
Semi-natural vegetation Light	-5,62	-26,45	21,10	0,649155698
Semi-natural vegetation Minimal	7,19	-13,81	33,32	0,532385086

Table 8-10 Estimates, Confidence Intervals and p-values of taxa richness results

Indicator	Mean Estimate (%)	Low CI Value (%)	High CI Value (%)	p-value
Values Figure 3-1				
Natural vegetation	0	0	0	-
Cropland Intense	-18,91	-30,69	-5,14	0,0088668
Cropland Light	-26,51	-39,28	-11,06	0,0015669
Cropland Minimal	-17,73	-28,75	-5,00	0,0078628
Pasture Intense	-15,86	-31,94	4,01	0,1105523
Pasture Light	-1,66	-11,20	8,90	0,7479220
Pasture Minimal	21,03	7,23	36,62	0,0020209
Semi-natural vegetation Intense	5,75	-11,45	26,28	0,5372699
Semi-natural vegetation Light	35,55	15,72	58,79	0,0001653
Semi-natural vegetation Minimal	19,93	5,06	36,91	0,0071592
Values Figure 3-2				
Mean temperature	-9,83	-23,76	6,65	0,2270551
Crop diversity	-7,18	-13,26	-0,68	0,0310267
Natural habitat cover	-4,55	-7,48	-1,52	0,0035095
Small field prevalence	14,78	5,62	24,72	0,0011596
Pesticide use	-7,53	-11,26	-3,64	0,0001953
Urban cover	-0,29	-2,41	1,87	0,7875193
Large field prevalence	0,37	-2,32	3,13	0,7899777
Water cover	-0,27	-4,00	3,61	0,8912463
Mean temperature	-9,83	-23,76	6,65	0,2270551
Values Figure 3-3 – Crop diversity				
Natural vegetation	-7,18	-13,26	-0,68	0,030942128
Cropland Intense	16,32	8,10	25,17	5,28052E-05
Cropland Light	14,27	2,91	26,89	0,01251147
Cropland Minimal	-0,96	-9,12	7,94	0,825989889
Pasture Intense	-7,85	-21,91	8,75	0,333518485
Pasture Light	10,92	2,23	20,35	0,012777853
Pasture Minimal	7,96	0,40	16,10	0,038695764
Semi-natural vegetation Intense	0,85	-8,95	11,71	0,871257131
Semi-natural vegetation Light	-8,72	-17,96	1,55	0,093450958
Semi-natural vegetation Minimal	-3,24	-9,66	3,64	0,346966953
Values Figure 3-3 – Small field prevalence				
Natural vegetation	14,78	5,62	24,72	0,001152975
Cropland Intense	3,61	-4,15	12,01	0,372030948
Cropland Light	4,23	-3,59	12,69	0,298034899
Cropland Minimal	-1,27	-11,63	10,31	0,821694848
Pasture Intense	10,89	-3,02	26,79	0,130785992
Pasture Light	2,13	-5,65	10,54	0,60206331
Pasture Minimal	-7,54	-15,02	0,59	0,068413249
Semi-natural vegetation Intense	-0,77	-10,85	10,45	0,88794784
Semi-natural vegetation Light	-8,33	-18,25	2,79	0,13650913
Semi-natural vegetation Minimal	-7,67	-16,05	1,55	0,10035245

9. Supplementary information and code availability

As complement to this work, the following supplementary information is attached to the report:

- **S1:** Detail description of issues found in the primary literature review of the dataset
- **S2:** Excel file with results of all ran models, including DHARMa and check_model() plots
- **S3:** Answers to questionnaire to identify issues in primary literature

Additionally, R scripts for the complete analyses were uploaded to GitHub and are available at https://github.com/cvattuonet/agri-arthropods-europe-predicts

10. Ethics Statement

Generative AI tools were used in the preparation of this report. GitHub Copilot was used for code competition and debugging while writing R scripts. Claude (model Opus 5) was mainly used to edit sections of the text being drafted on my own, for explaining statistical concepts and for code generation to improve plotted figures as well as debugging. All research decisions are of my own and the supervisory team, including study design, curation of the datasets, review of primary literature behind it, selection and use of spatial products and predictors, modelling choices and interpretation of results. Bibliographic review was done without AI tools, so every source cited was screened or read in detailed by me.